# The Cerebrum and Cerebral Cortex

Department Anatomy

Neuroanatomy (2)

First Part

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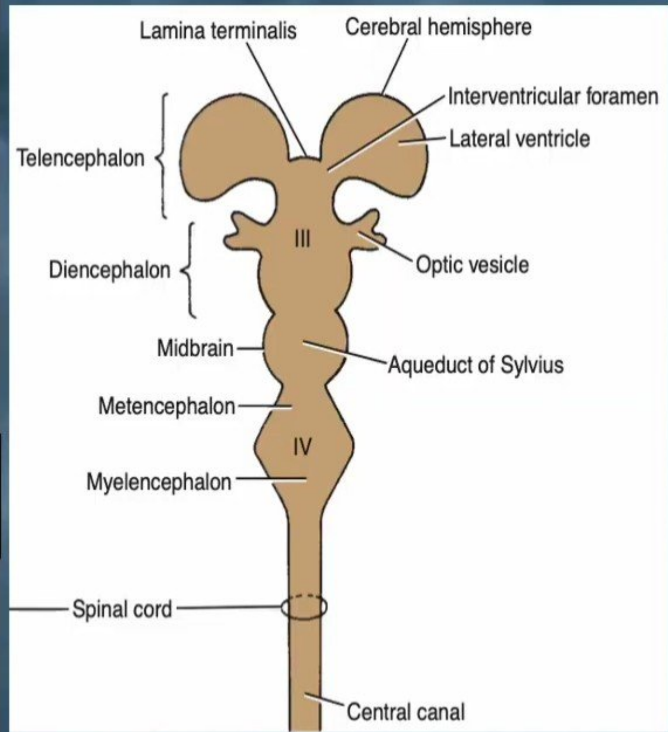
# lecture Points:

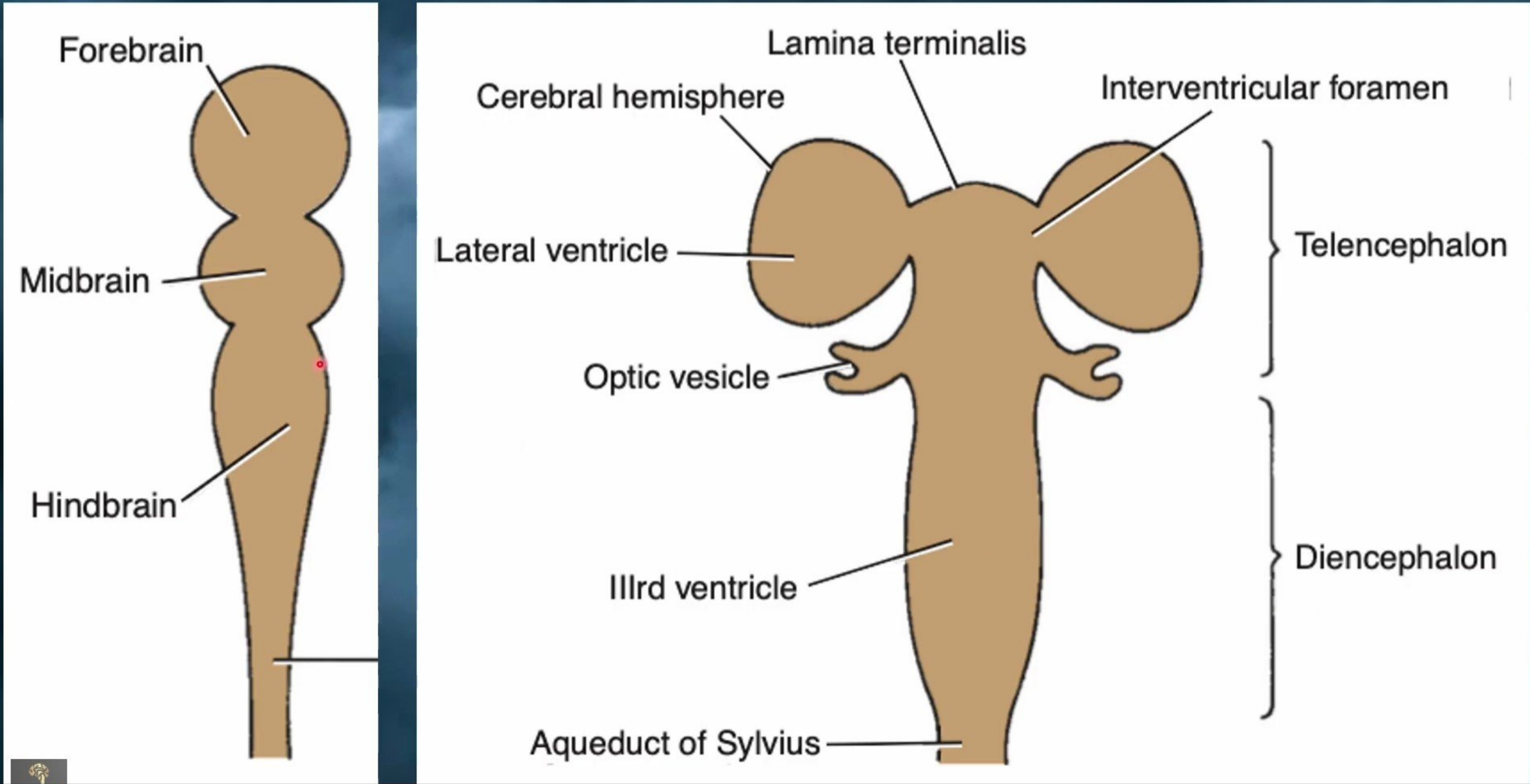
- Embryonic (developmental) divisions of the Brain.
- Division of the forebrain vesicle into the telencephalon and the diencephalon.
- The basic components of the CNS.
- Cerebrum – Cerebral Hemisphere
- Structure of the cerebral cortex.
- Layers of the Cerebral Cortex.
- Nerve Fibers of the Cerebral Cortex.
- Cerebral Features. (The cerebral hemispheres)
- Lobes of the Cerebral Cortex. (Sulci , Gyri , Functional Areas & Lesions)
  - Frontal lobe.
  - Parietal lobe.
  - Temporal Lobe.
  - Occipital lobe.
- Insula.
- *Limbic Lobe (The limbic system).*
- Basal Nuclei.
- White Matter of the Cerebral Hemispheres.
- Lobes of Cerebral Cortex and Associated Integration.
- Motor homunculus on the precentral gyrus.
- Functional Areas of the Cerebral Cortex
- Domenet hemospher.
- Association areas.
- Probable nerve pathways involved in reading a sentence and repeating it out.
- Probable nerve pathways involved with hearing a question and answering it.

# Embryonic (developmental) divisions of the Brain

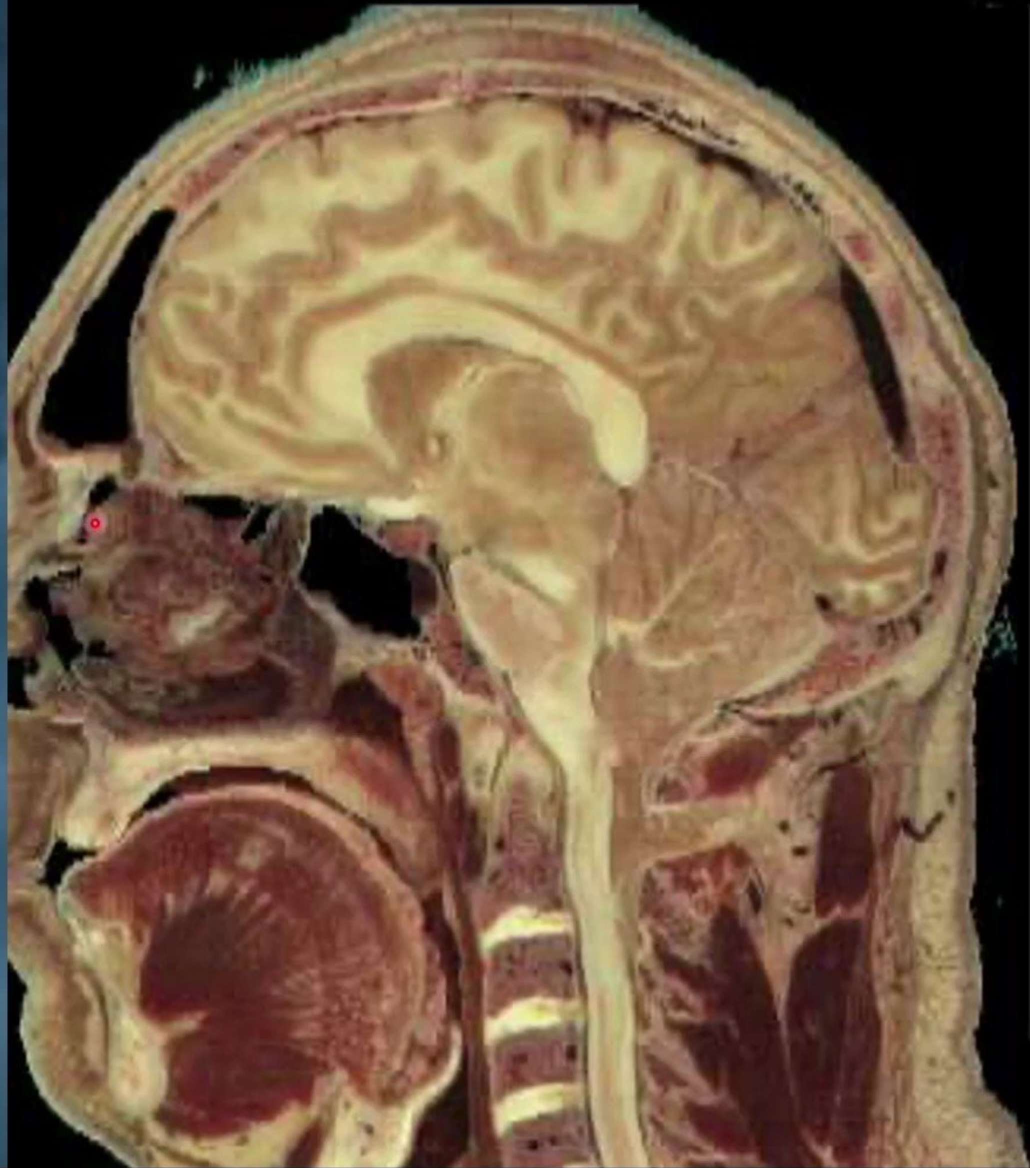
| Primary vesicle | Secondary vesicle | Derivatives |
|---|---|---|
| Prosencephalon | Telencephalon | Cerebral cortex<br>Cerebral white matter<br>Basal ganglia |
| | Diencephalon | Thalamus<br>Hypothalamus<br>Subthalamus<br>Epithalamus |
| Mesencephalon | Mesencephalon | Midbrain |
| Rhombencephalon | Metencephalon | Cerebellum<br>Pons |
| | Myelencephalon | Medulla oblongata |



# Division of the forebrain vesicle into the telencephalon and the diencephalon

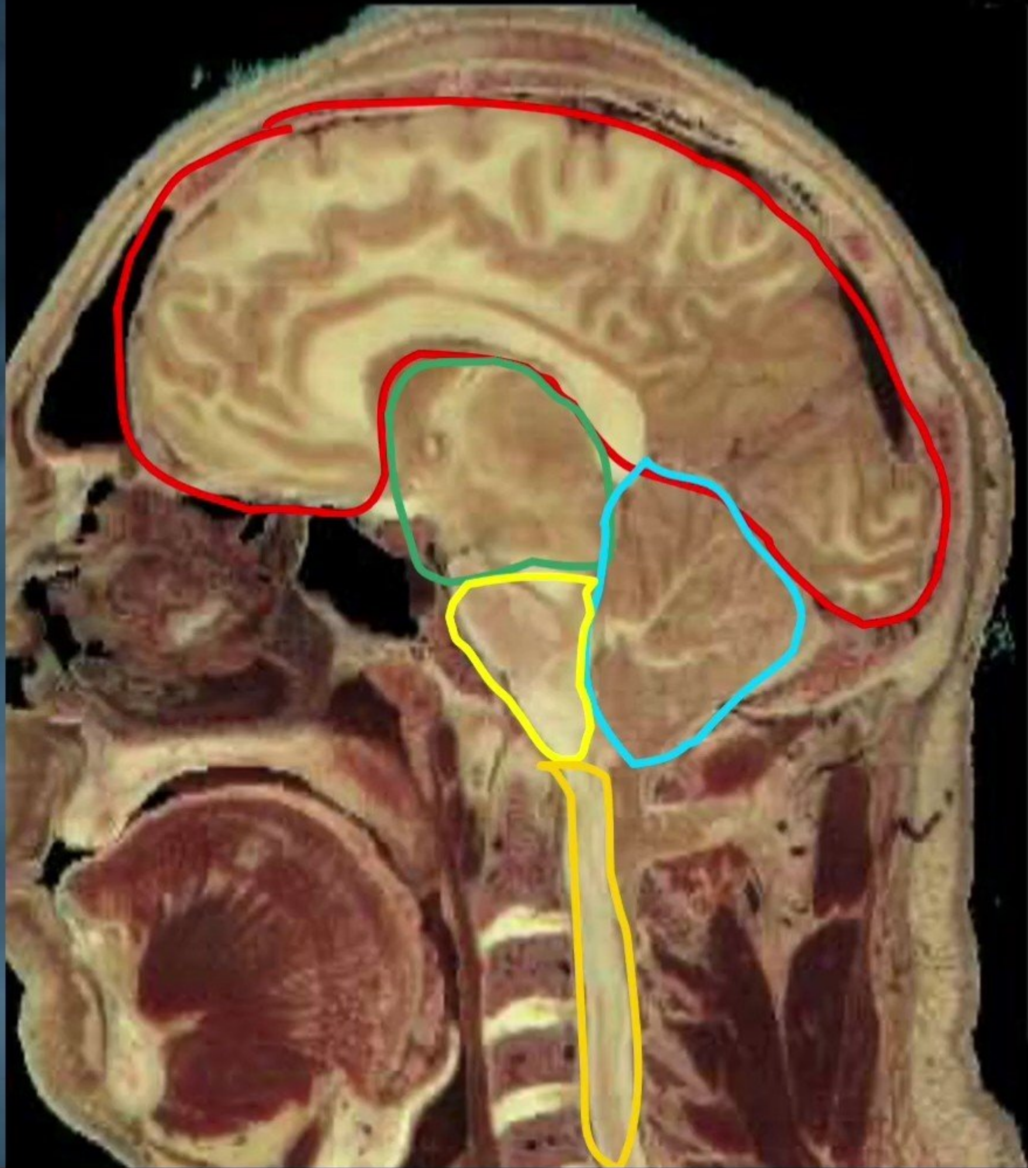


The basic components of the CNS include the:



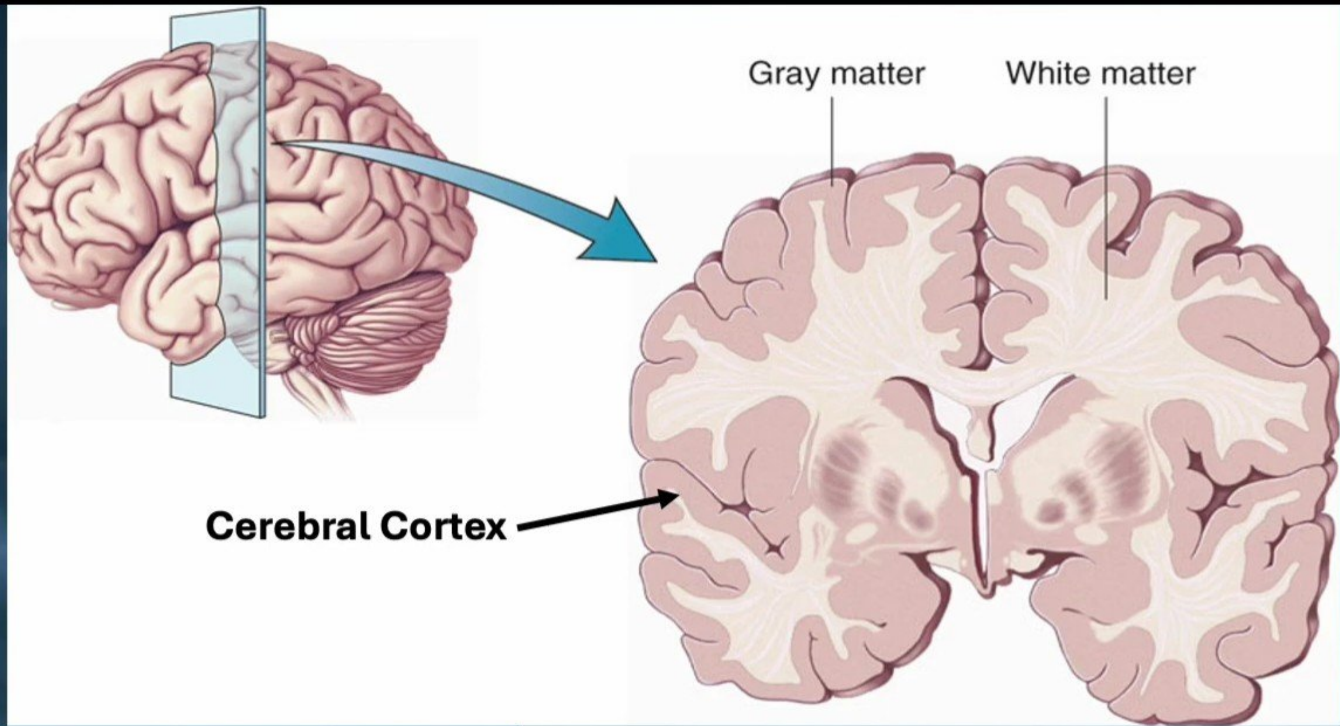
The basic components of the CNS include the:

- Cerebrum
- Diencephalon
- Cerebellum
- Brain stem
- Spinal cord



# Cerebrum – Cerebral Hemisphere

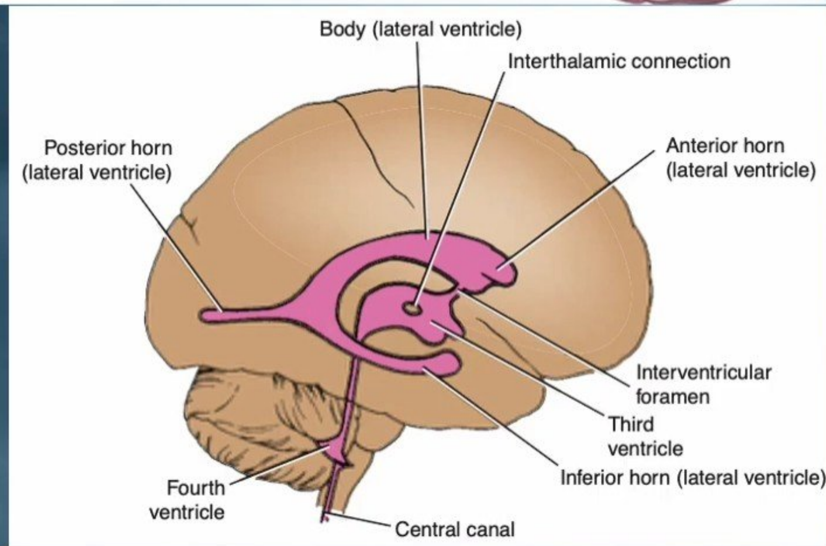
1) Cerebral Cortex. (lobes, gyri and sulci)
2) Lateral Ventricles.
3) Cerebral White Matter.
4) Basal Ganglia.
5) Thalamus.




## Where the Gray Matter??

### Cerebral Cortex:

**The outermost layer of gray matter making up the superficial aspect of the cerebrum.**



# Structure of the cerebral cortex

- It is composed of gray matter and contain 10 billion neurons.
- **The following types of nerve cells are present in the cerebral cortex:**

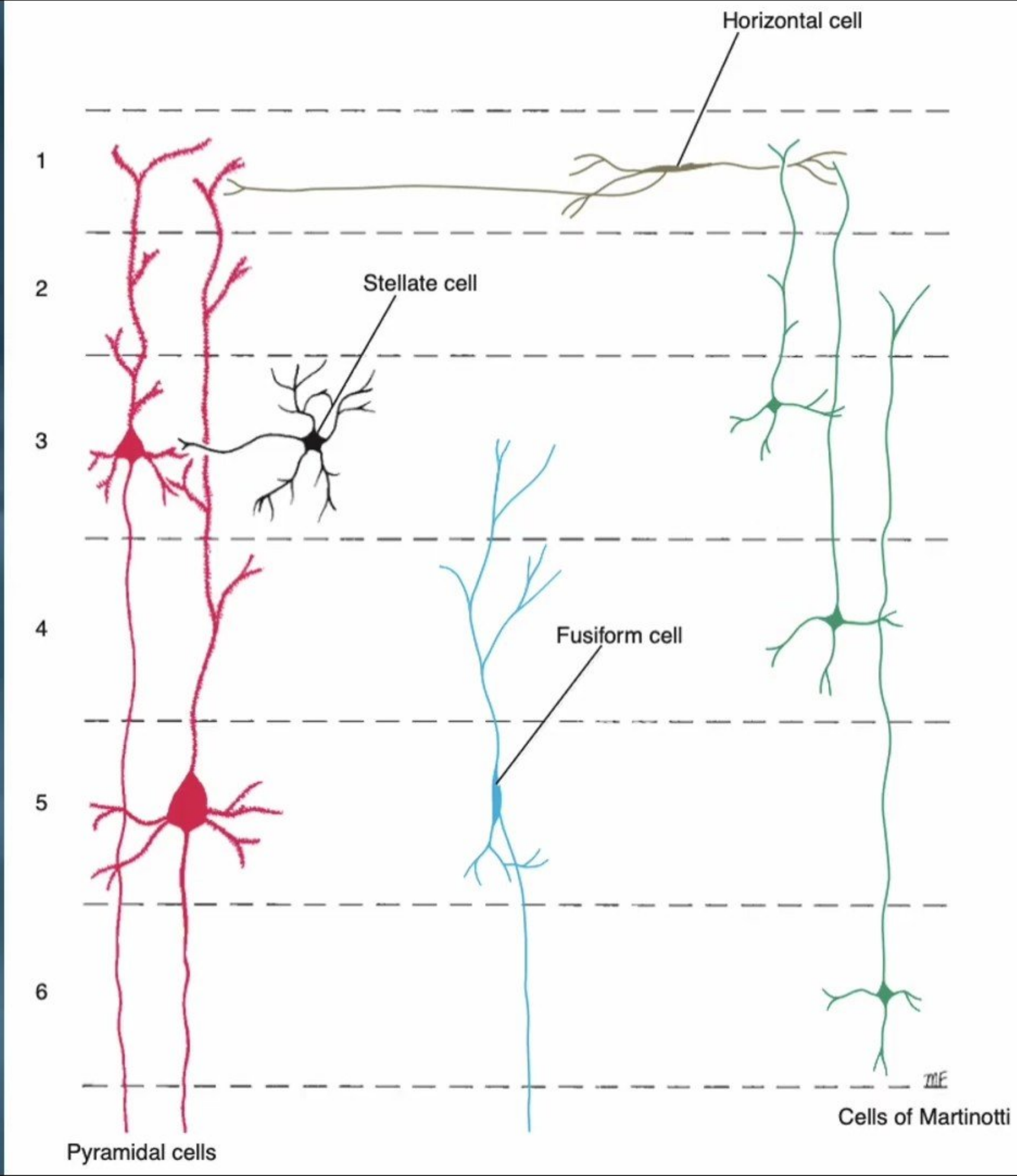
(1) Pyramidal cells.

(2) Stellate cells.

(3) Fusiform cells.

(4) Horizontal cells of Cajal.

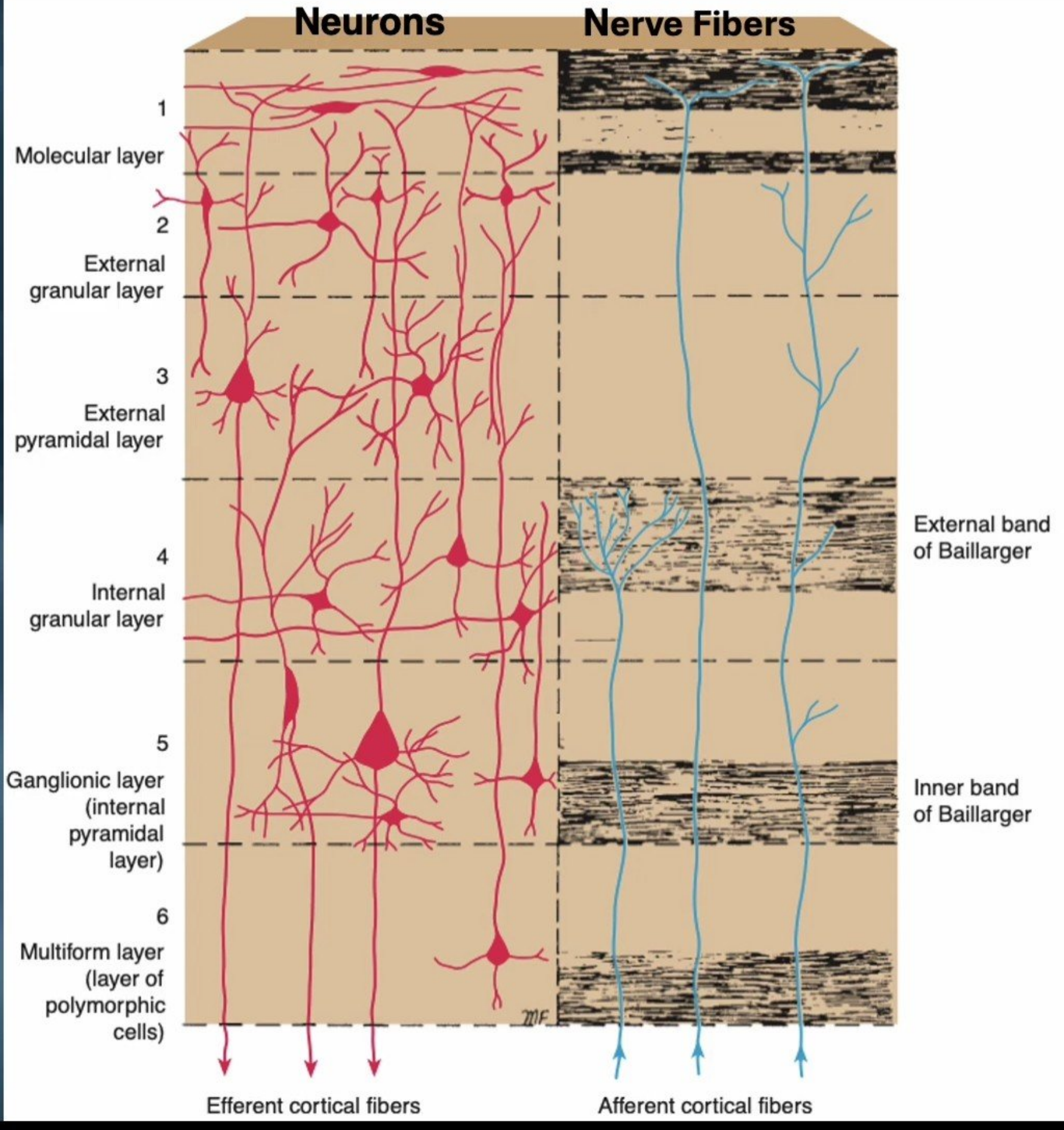
(5) Cells of Martinotti.



# Layers of the Cerebral Cortex

- It is convenient for descriptive. purposes, to divide the cerebral cortex into: layers that may be distinguished by the type, density and arrangement of their cells.

1) Molecular layer (plexiform layer).
2) External granular layer.
3) External pyramidal layer.
4) Internal granular layer.
5) Ganglionic layer (internal pyramidal layer).
6) Multiform layer (layer of polymorphic cells).

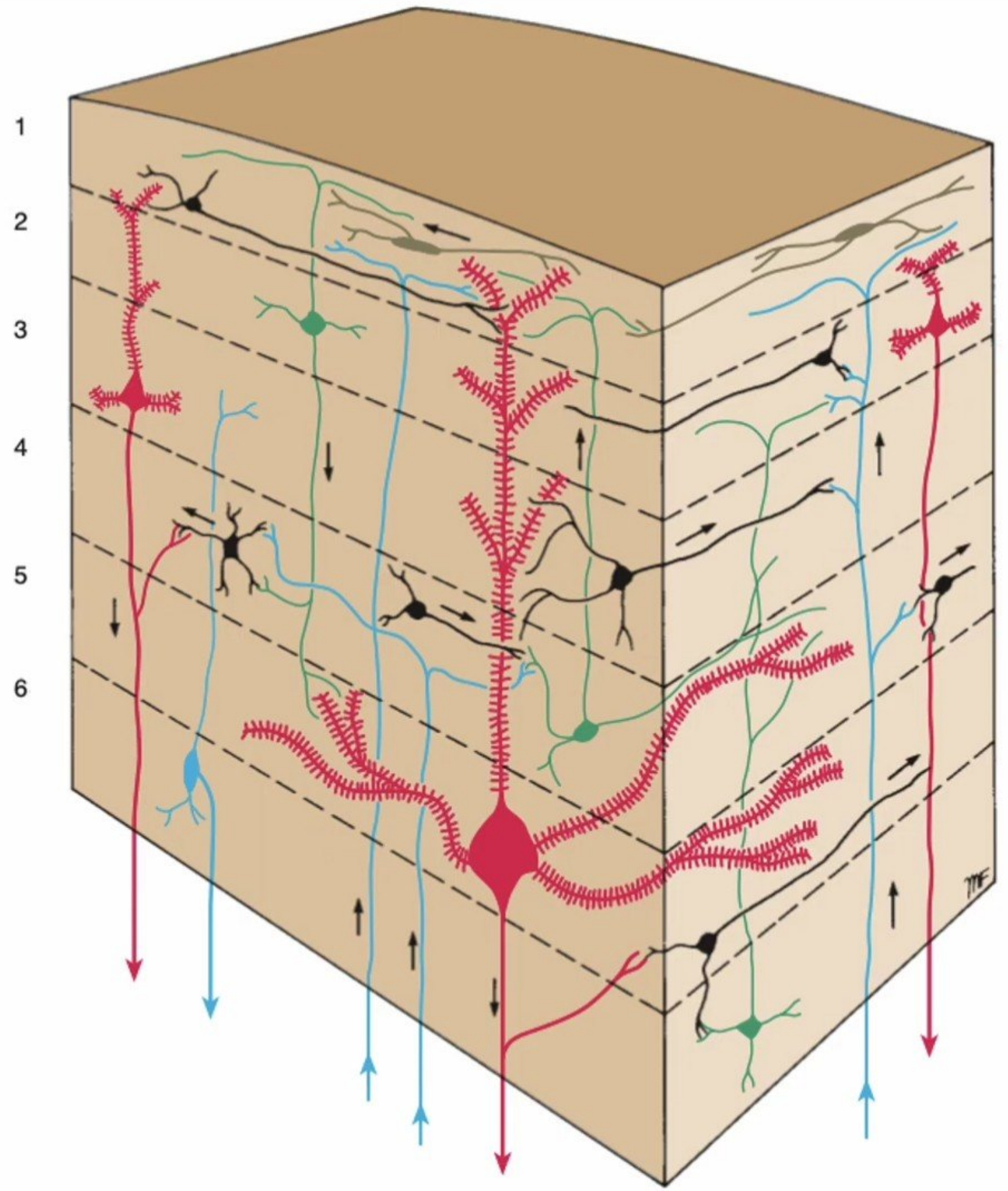


# Nerve Fibers of the Cerebral Cortex

The nerve fibers of the cerebral cortex are arranged both **radially** and **tangentially.**

## The radial fibers:

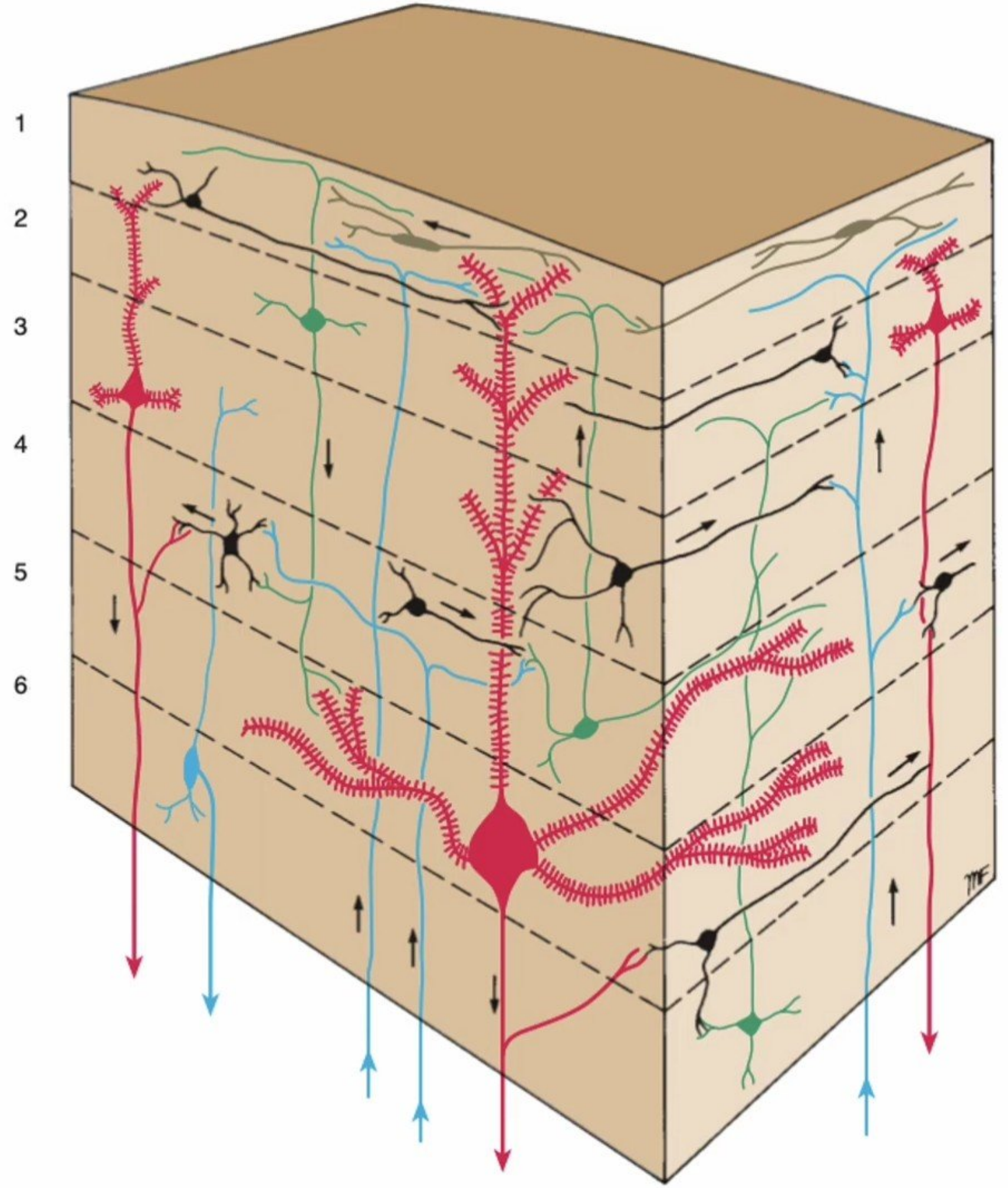
- run at right angles to the cortical surface.
- They include:
  - ✓ The afferent entering projection, association, and commissural fibers, (which terminate within the cortex, and
  - ✓ The axons of pyramidal, stellate, and fusiform cells, which leave the cortex to become projection, association, and commissural fibers of the white matter of the cerebral hemisphere.



# Structure of the cerebral cortex

## The tangential fibers

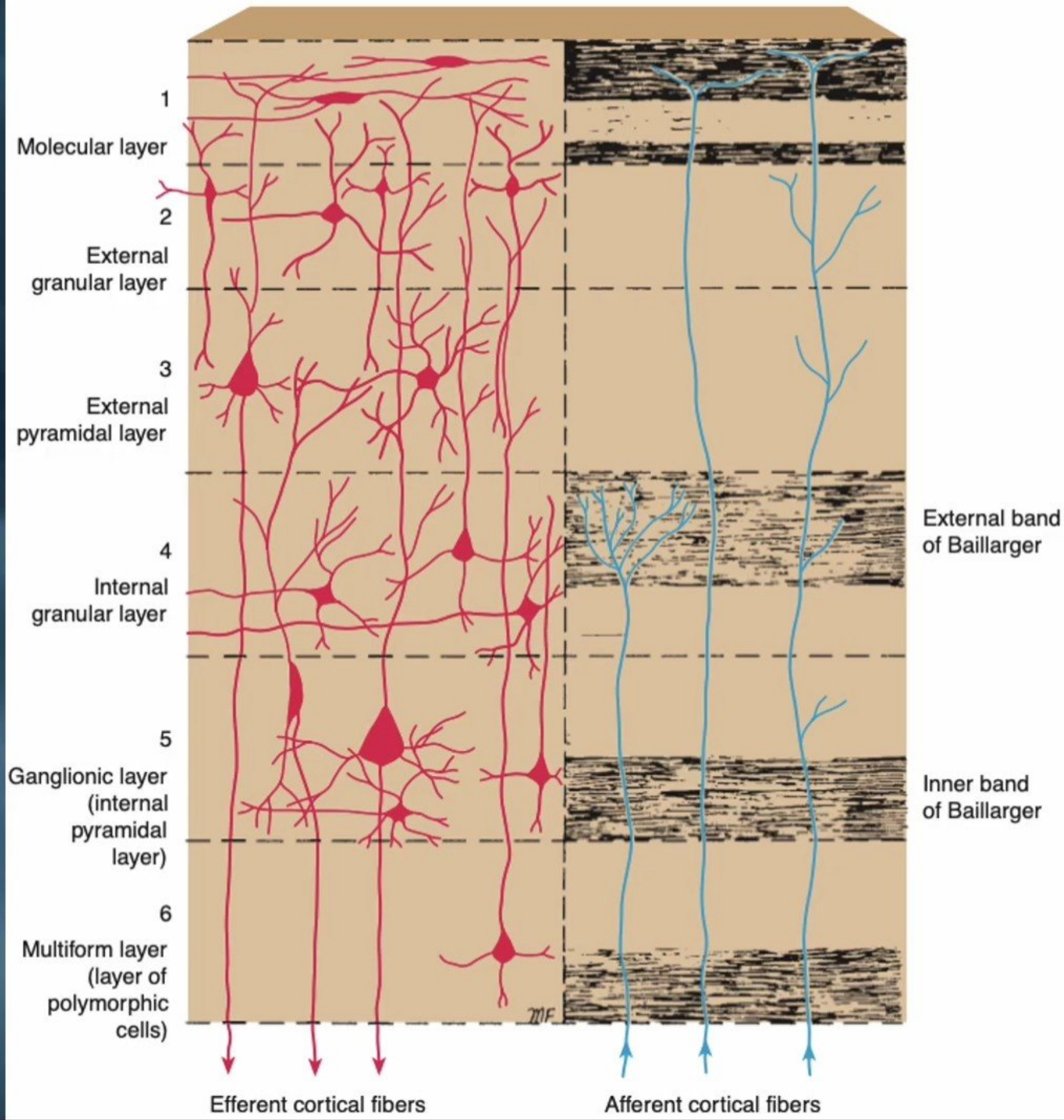
- They are run parallel to the cortical surface.
- The collateral and terminal branches of afferent fibers.
- They also include the axons of horizontal and stellate cells and collateral branches of pyramidal and fusiform cells.



# Structure of the cerebral cortex

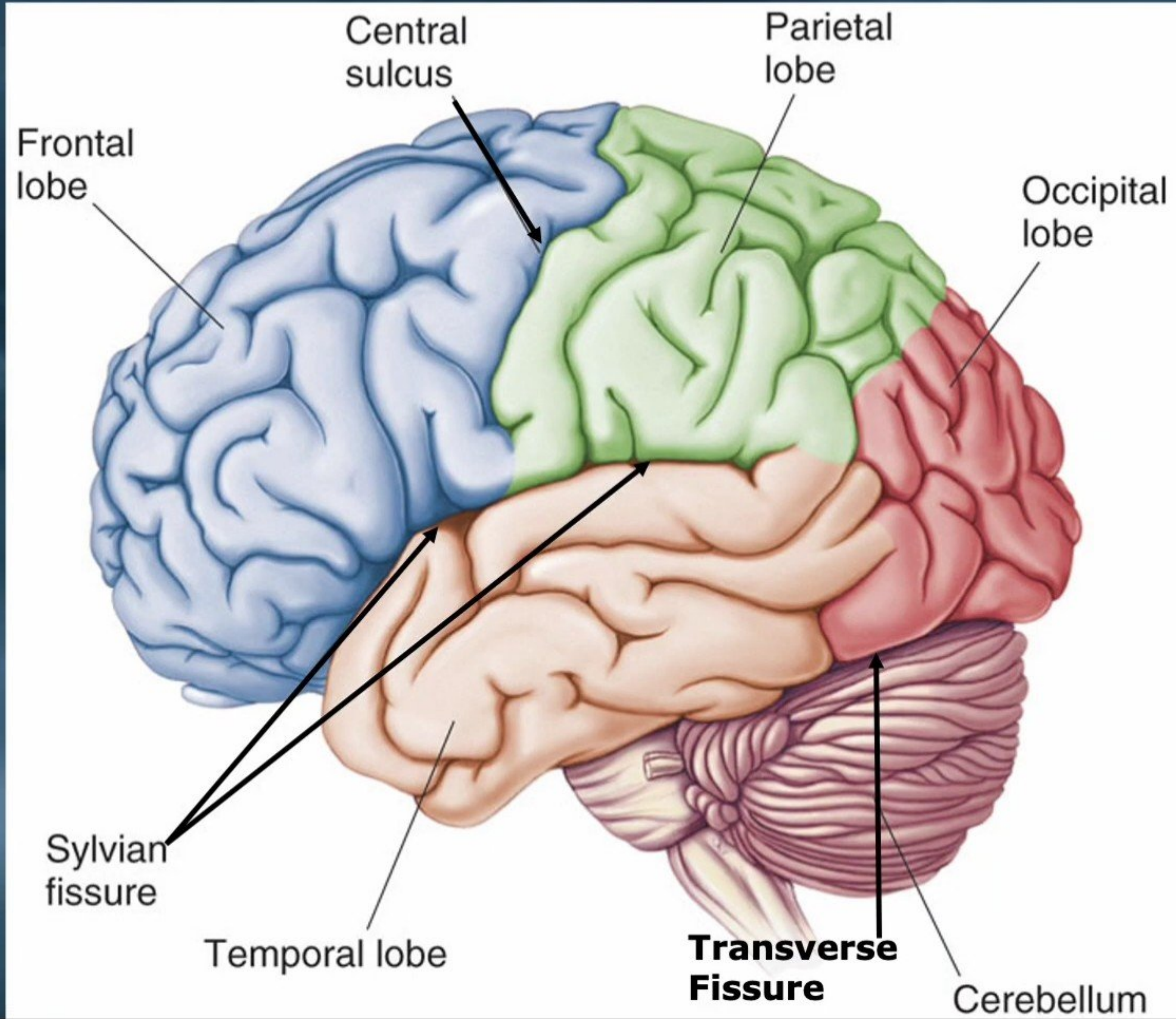
## The Tangential Fibers:

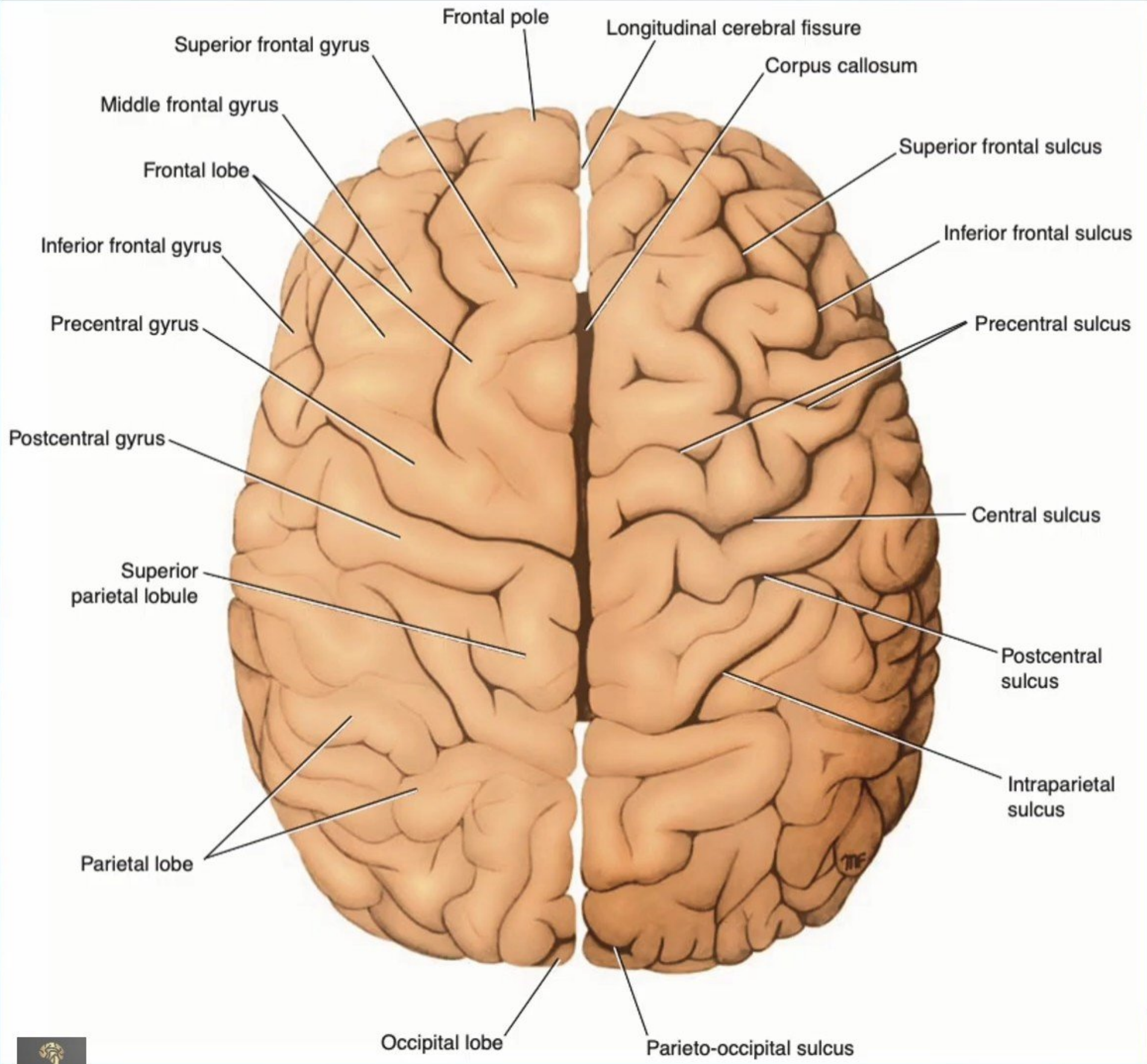
- They concentrated fibers.
- They are referred to as the outer and **inner bands of Baillarger,.**
- The bands of Baillarger are developed in the sensory areas due to the high concentration of the terminal parts of the **thalamocortical fibers.**
- In the **visual cortex**, the outer band of Baillarger is known as the stria of Gennari.
- Because of this obvious band, or stria, the visual cortex in the walls of the calcarine sulcus is sometimes called the **striate cortex**.



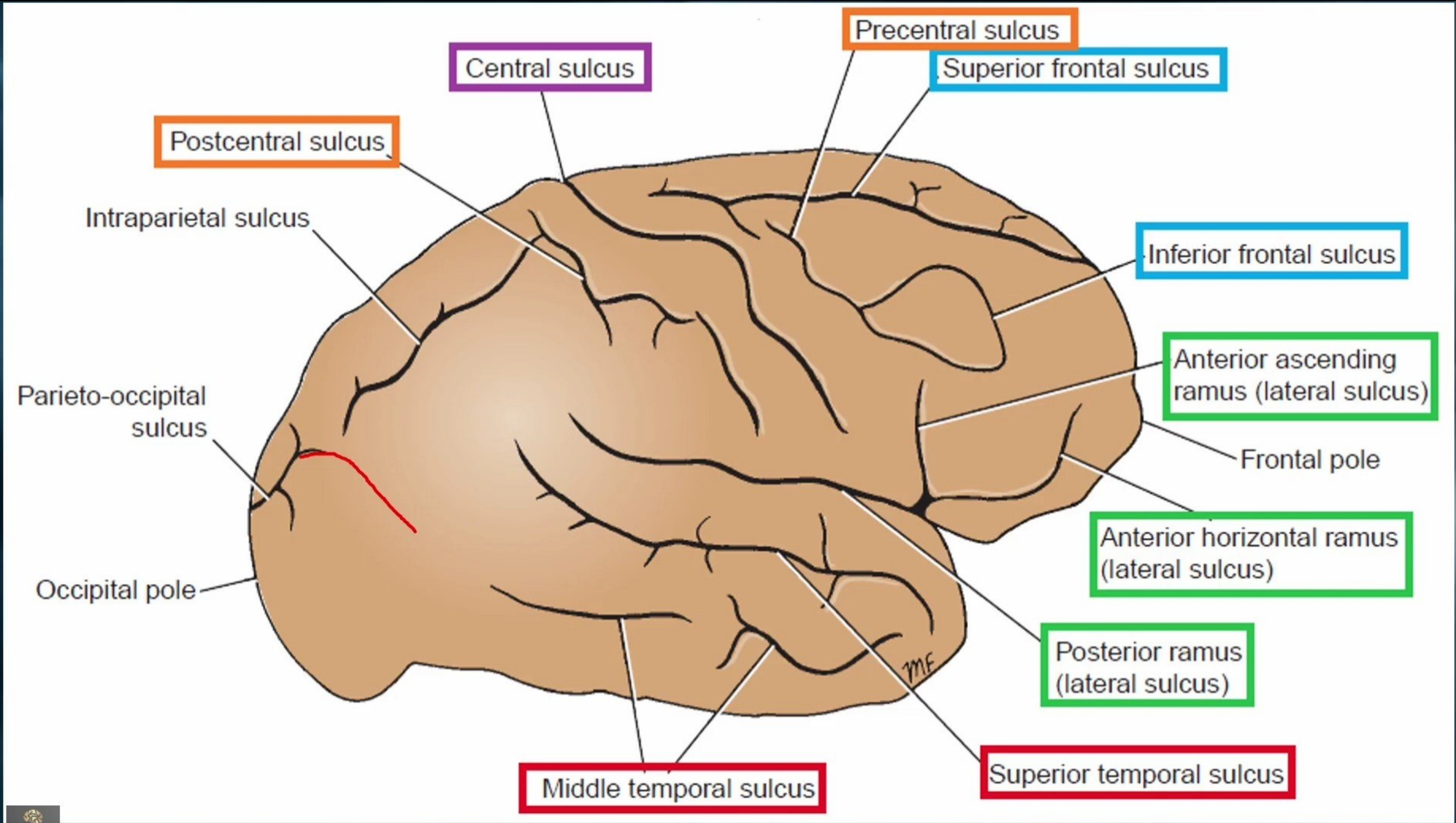
# Cerebral Features

- **Gyri** – Elevated ridges "winding" around the brain.
- **Sulci** – Small grooves dividing the gyri.
  - ✓ Central Sulcus – Divides the Frontal Lobe from the Parietal Lobe.'
  - ✓ Parieto-occipital sulci
  - ✓ Calcarine sulcus
- **Fissures** – Deep grooves, generally dividing large regions/lobes of the brain.
  - ✓ **Longitudinal Fissure** – Divides the two Cerebral Hemispheres.
  - ✓ **Transverse Fissure** – Separates the Cerebrum from the Cerebellum.
  - ✓ **Sylvian/Lateral Fissure** (lateral sulcus) – Divides the Temporal Lobe from the Frontal and Parietal Lobes.

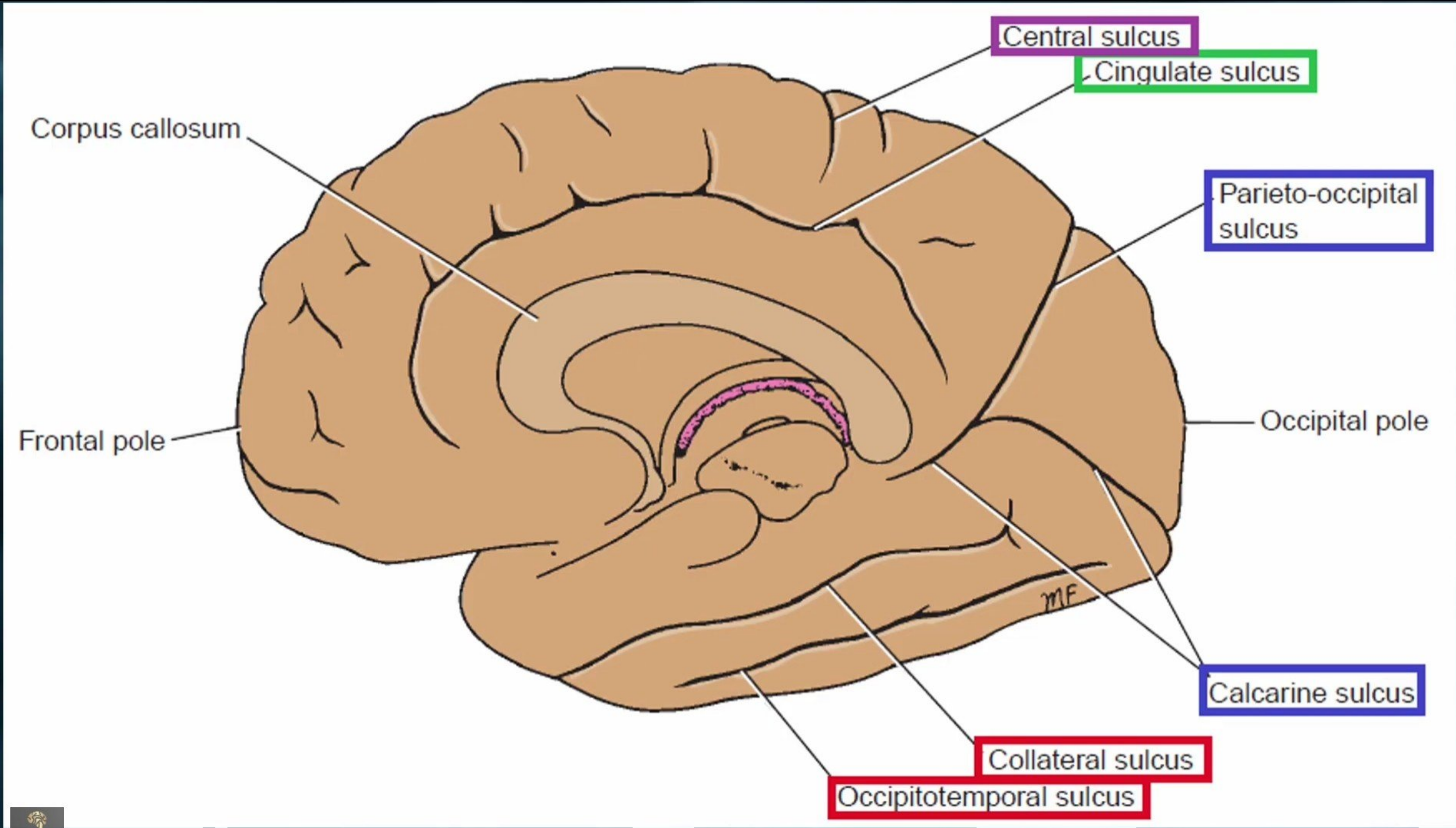



Frontal pole
Longitudinal cerebral fissure
Superior frontal gyrus
Corpus callosum
Middle frontal gyrus
Superior frontal sulcus
Frontal lobe
Inferior frontal sulcus
Inferior frontal gyrus
Precentral gyrus
Precentral sulcus
Postcentral gyrus
Central sulcus
Superior
parietal lobule
Postcentral
sulcus
Intraparietal
sulcus
Parietal lobe
Occipital lobe
Parieto-occipital sulcus


Longitudinal Fissure
Superior view of the cerebral hemispheres



Precentral sulcus
Central sulcus
Superior frontal sulcus
Postcentral sulcus
Intraparietal sulcus
Inferior frontal sulcus
Anterior ascending ramus (lateral sulcus)
Parieto-occipital sulcus
Frontal pole
Anterior horizontal ramus (lateral sulcus)
Occipital pole
Posterior ramus (lateral sulcus)
Middle temporal sulcus
Superior temporal sulcus

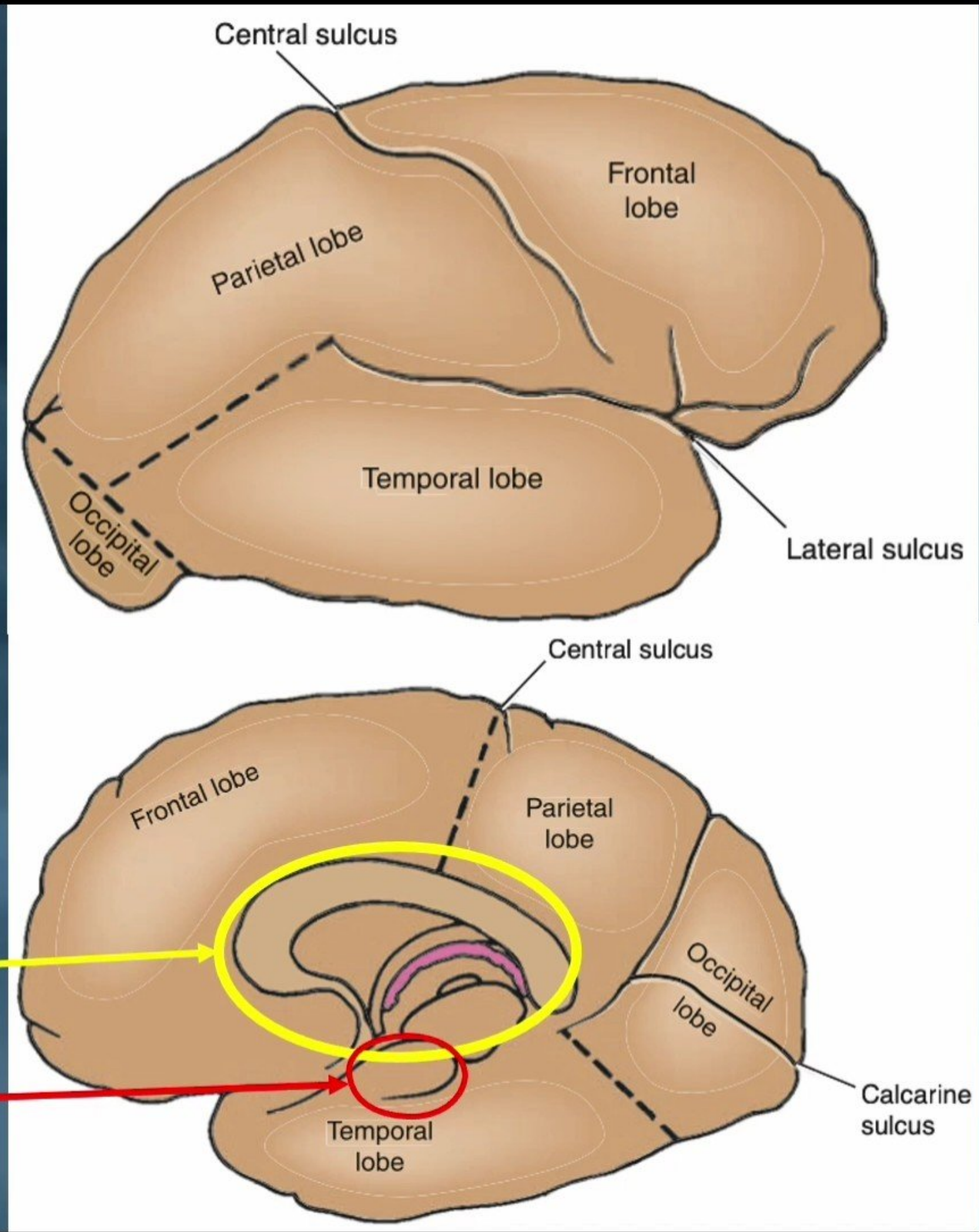


Central sulcus
Cingulate sulcus
Corpus callosum
Parieto-occipital sulcus
Frontal pole
Occipital pole
Calcarine sulcus
Collateral sulcus
Occipitotemporal sulcus

# The cerebral hemispheres

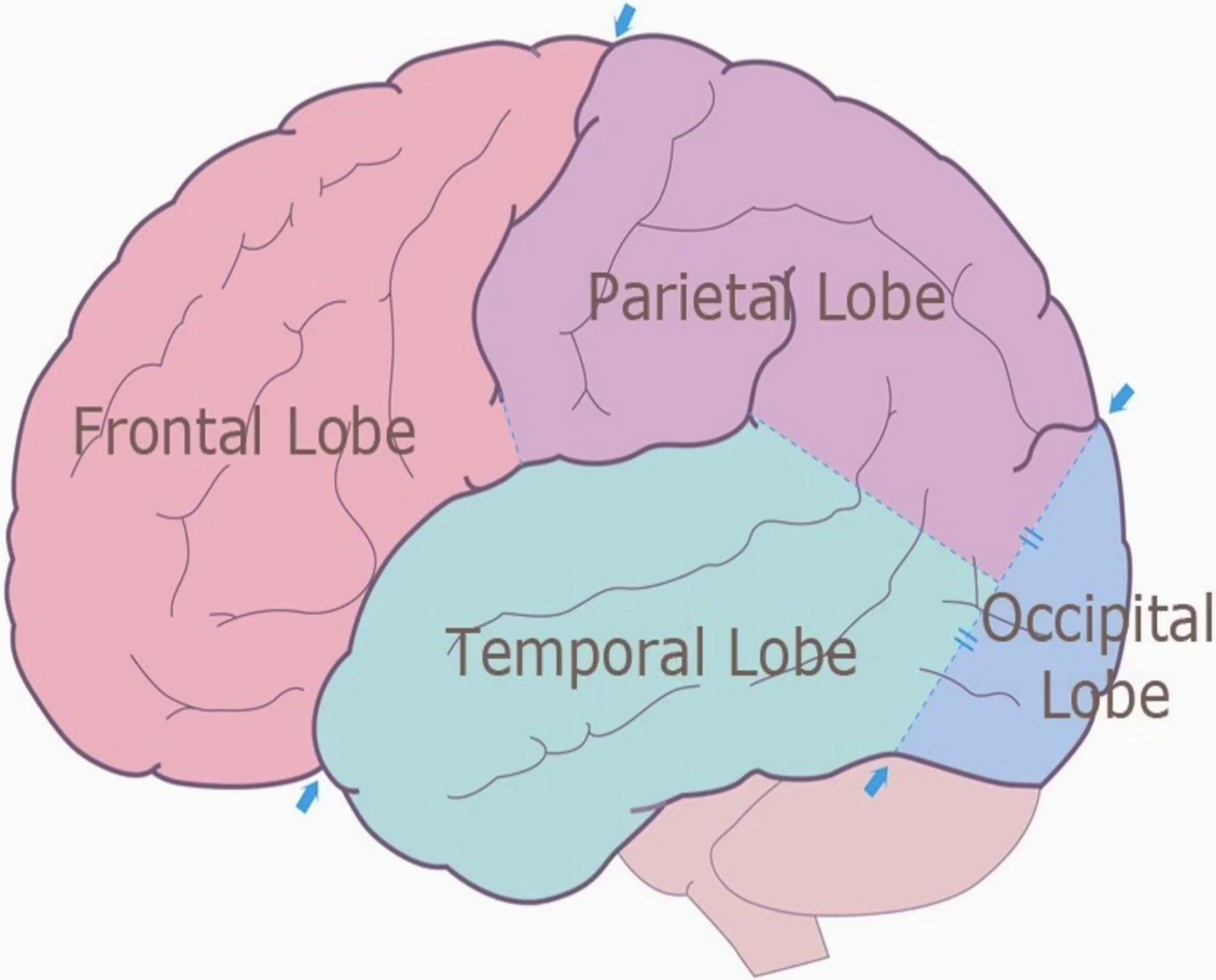
**The cerebral cortex consists of six lobes on each side:**

- **Frontal**
- **Parietal**
- **Occipital**
- **Temporal**
- Limbic
- Insular



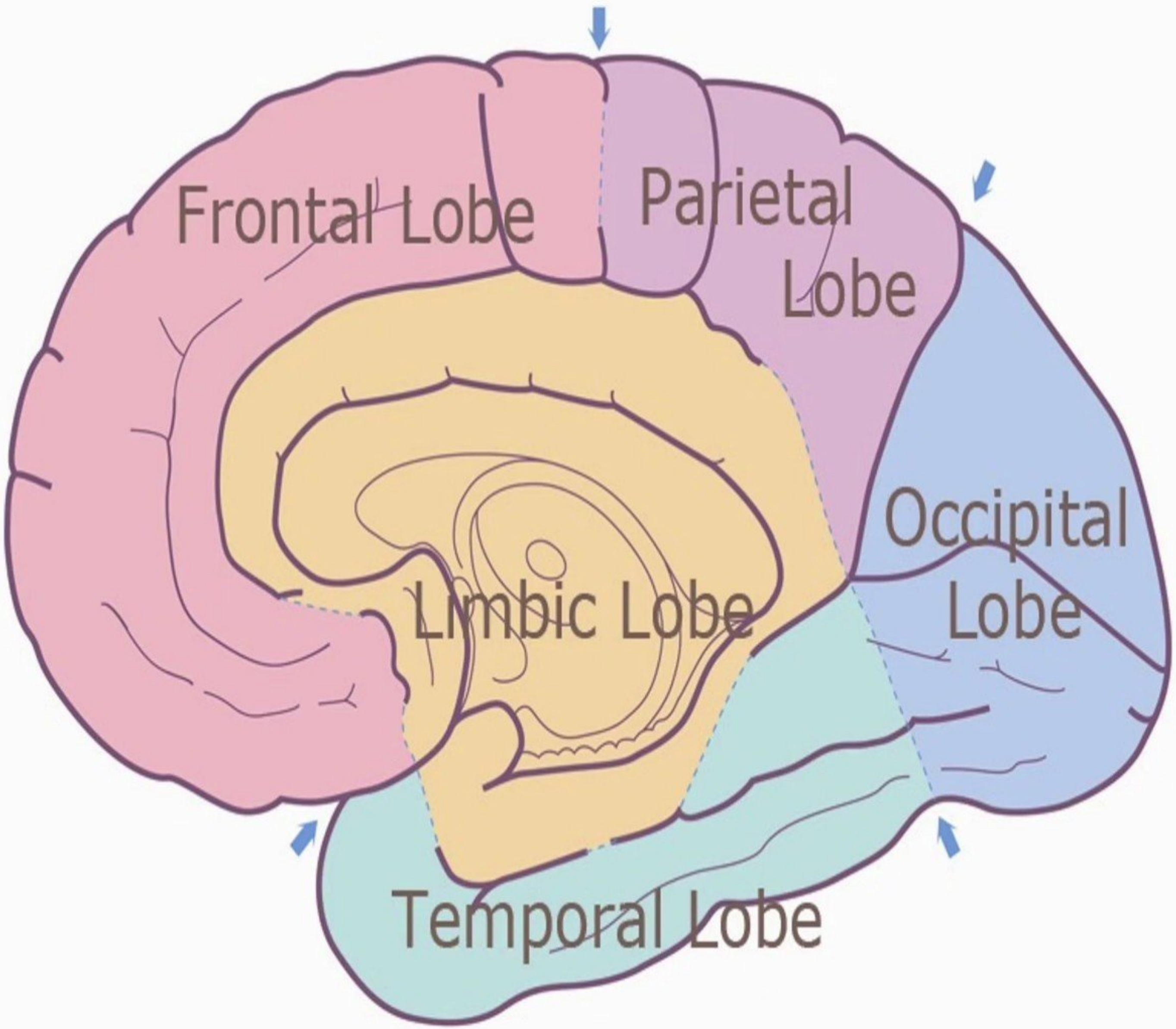
# Lobes of the Cerebral Cortex (lateral surface)

- ➢ Frontal lobe
- ➢ Parietal lobe
- ➢ Temporal Lobe
- ➢ Occipital lobe



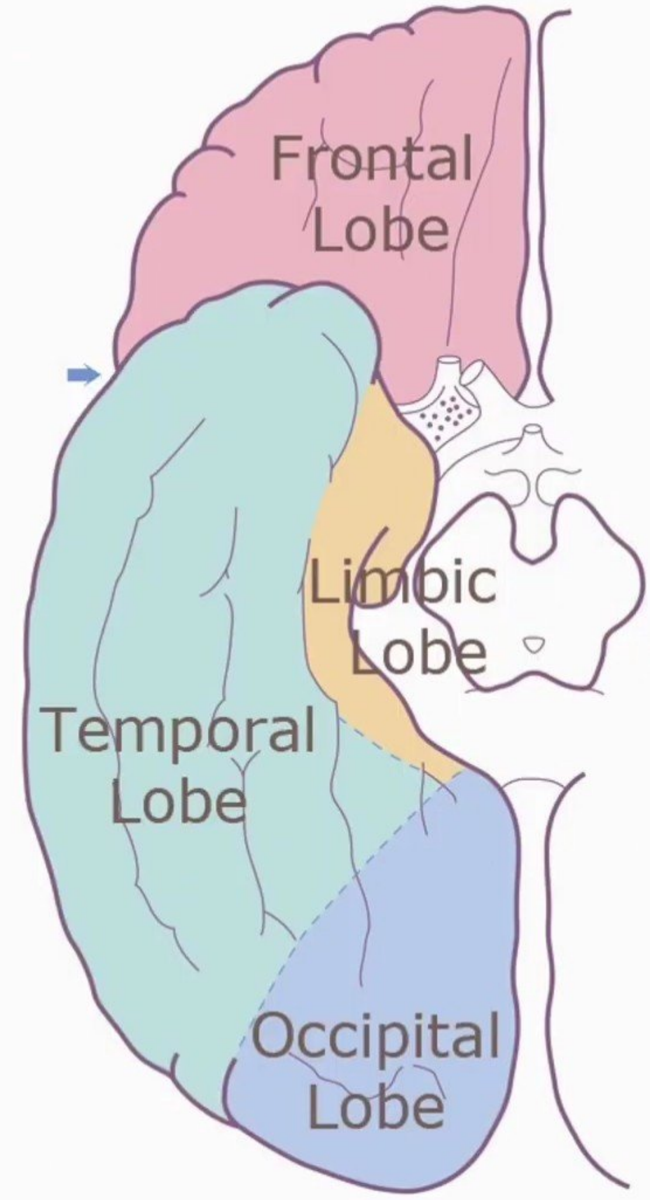
# Lobes of the Cerebral Cortex (medial surface)

- ➢ Frontal lobe.
- ➢ Parietal lobe.
- ➢ Temporal Lobe.
- ➢ Occipital lobe.
- ➢ Limbic lobe.



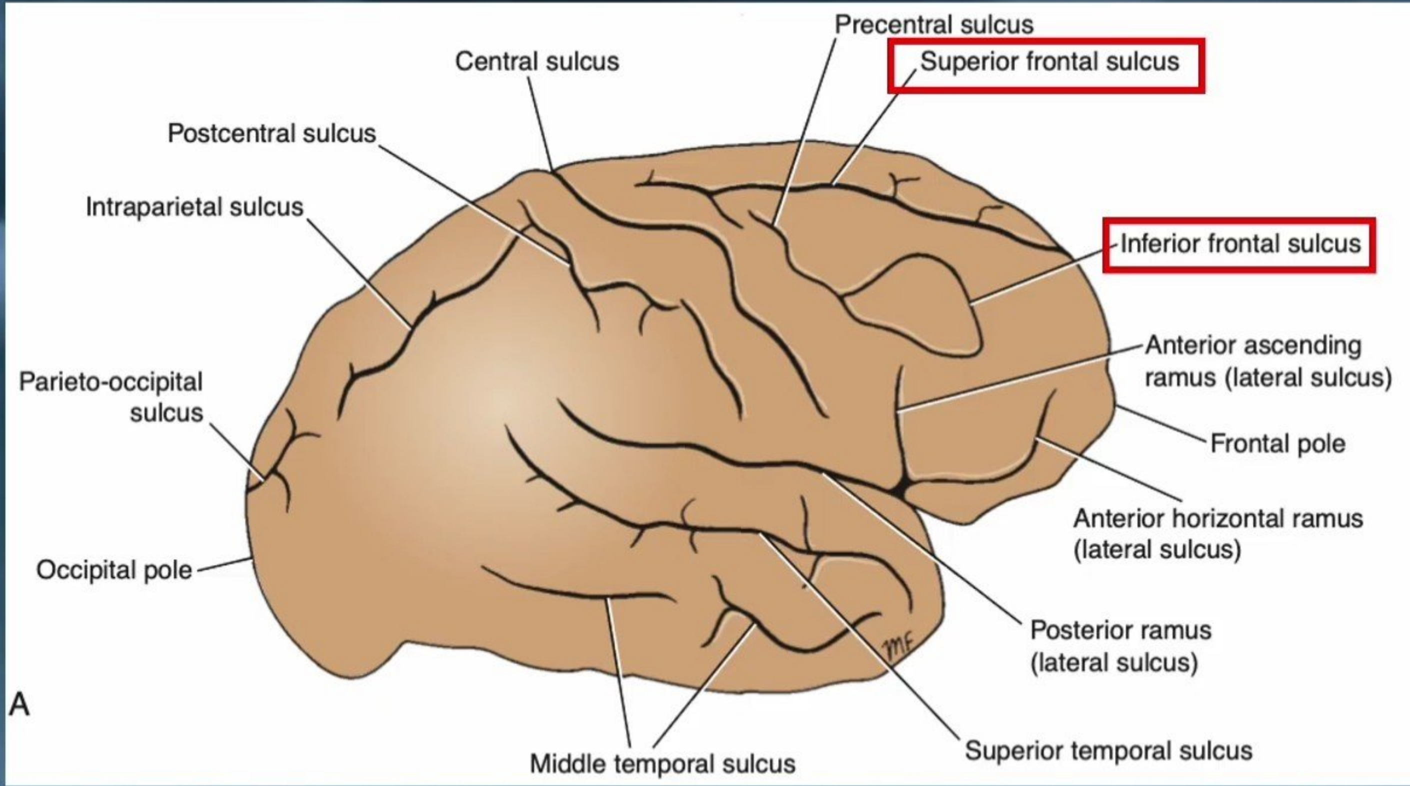
# Lobes of the Cerebral Cortex (basal surface)

- Frontal lobe
- Temporal Lobe
- Occipital lobe
- Limbic lobe



# Sulci of the Frontal Lobe

- The superior sulcus.
- The inferior frontal sulcus.



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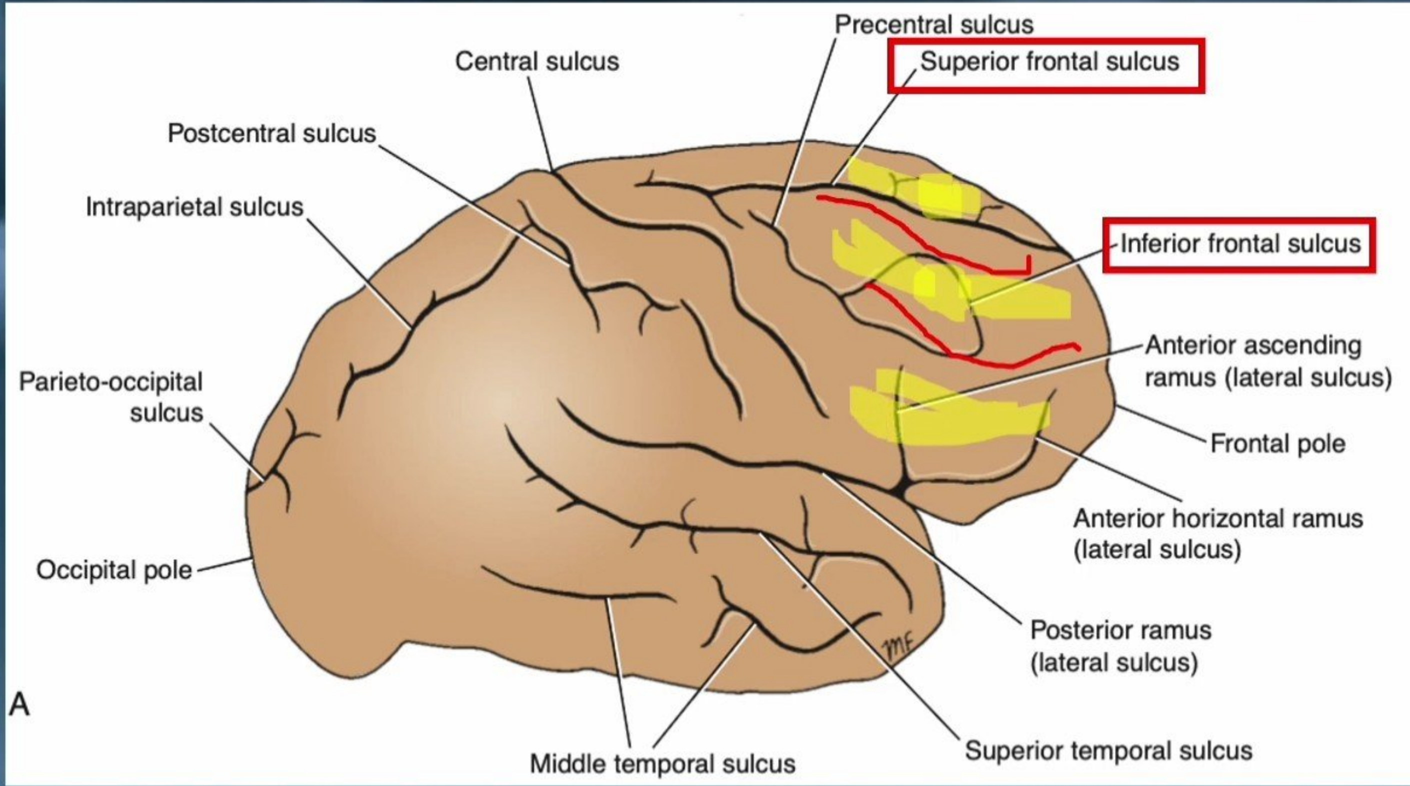
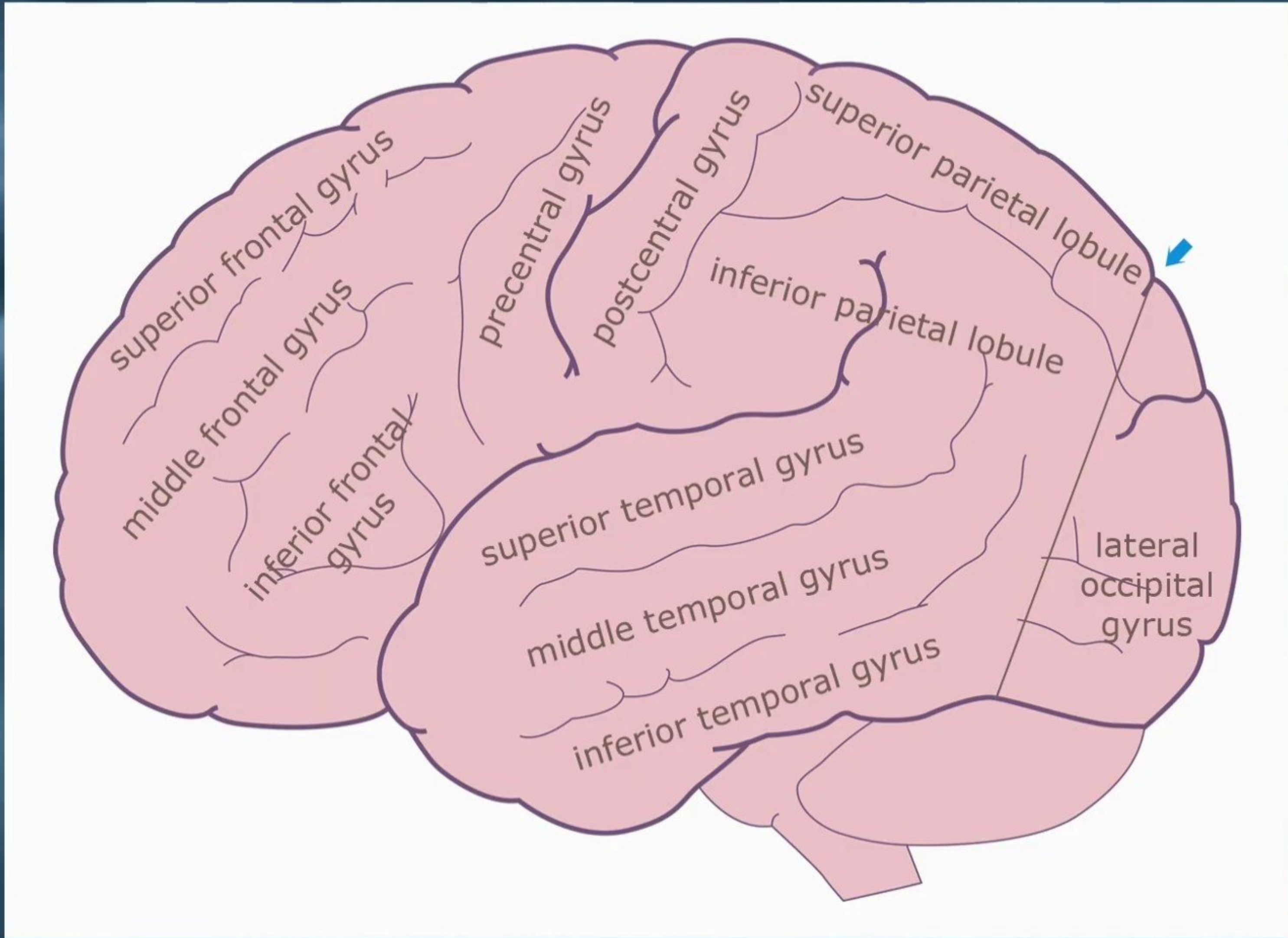


# *Gyri of the Frontal Lobe*

## Lateral Surface

- ✓ **Precentral Gyrus** – (primary motor area)
- ✓ **Superior Frontal Gyrus**
- ✓ **Middle Frontal Gyrus**
- ✓ **Inferior Frontal Gyrus** – (Broca's area)



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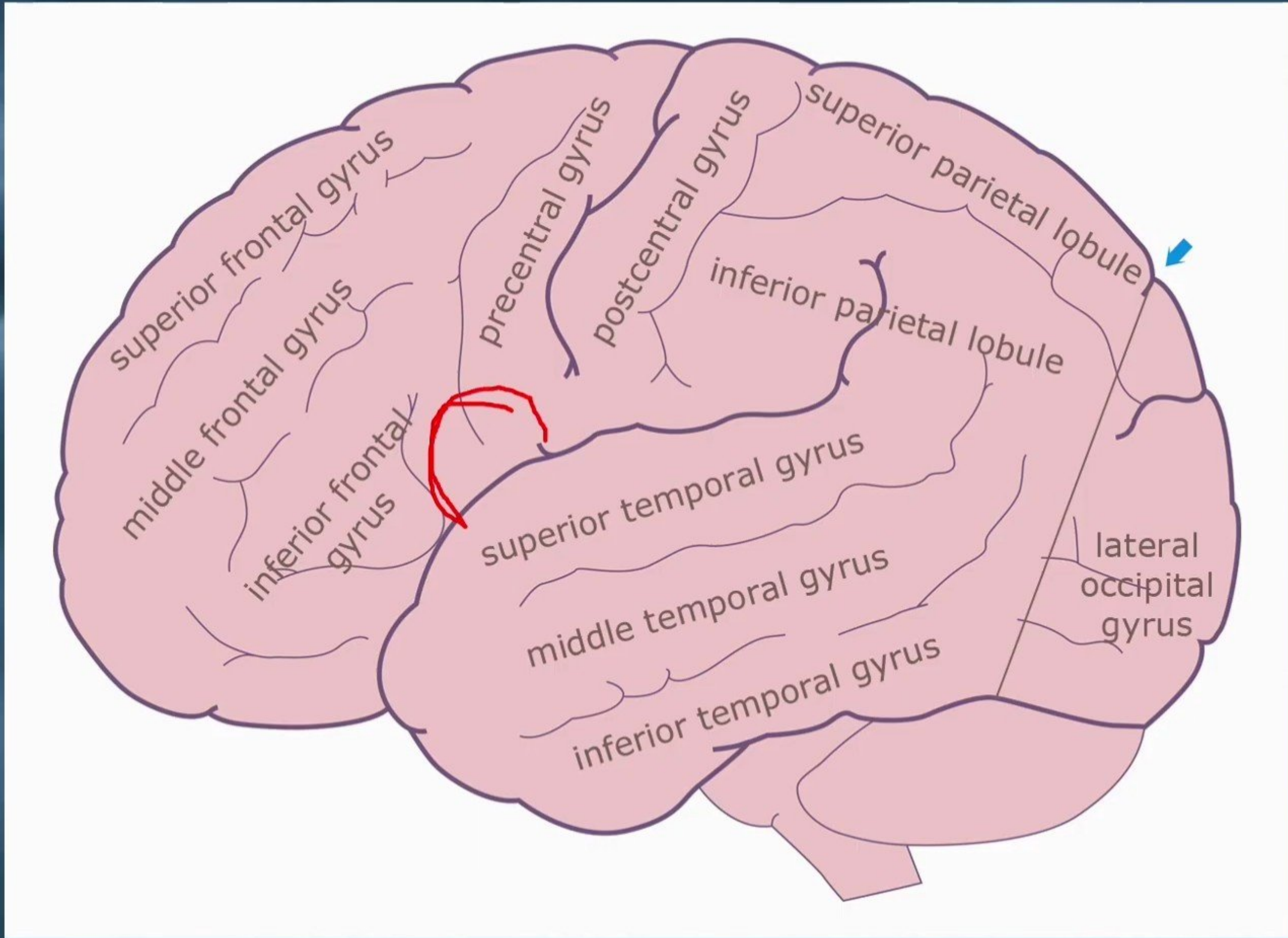
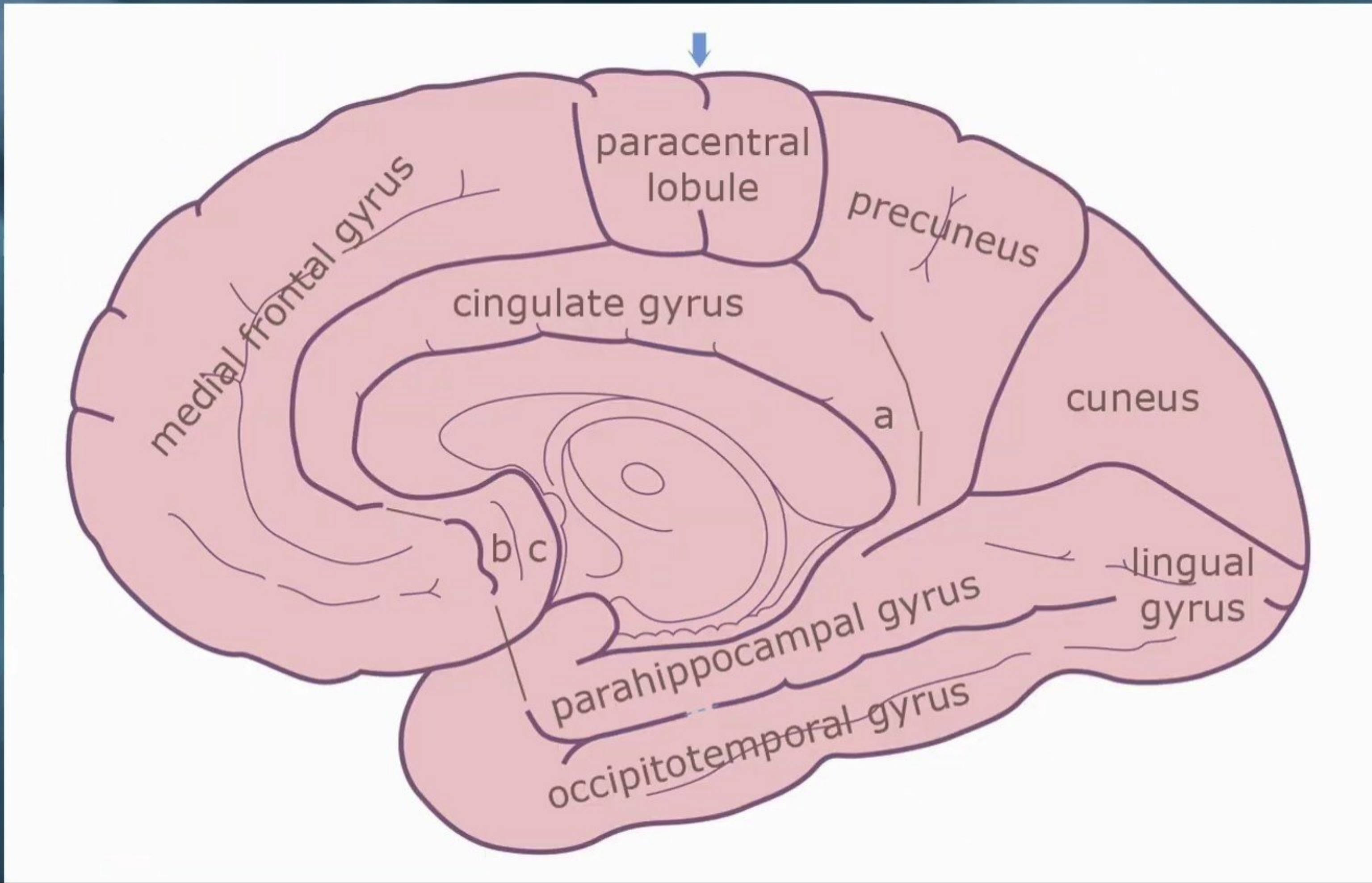


# *Gyri of the Frontal Lobe*

## Medial Surface

- ✓ **Medial Frontal Gyrus**
- ✓ **Paracentral Lobule** – ½ primary motor area



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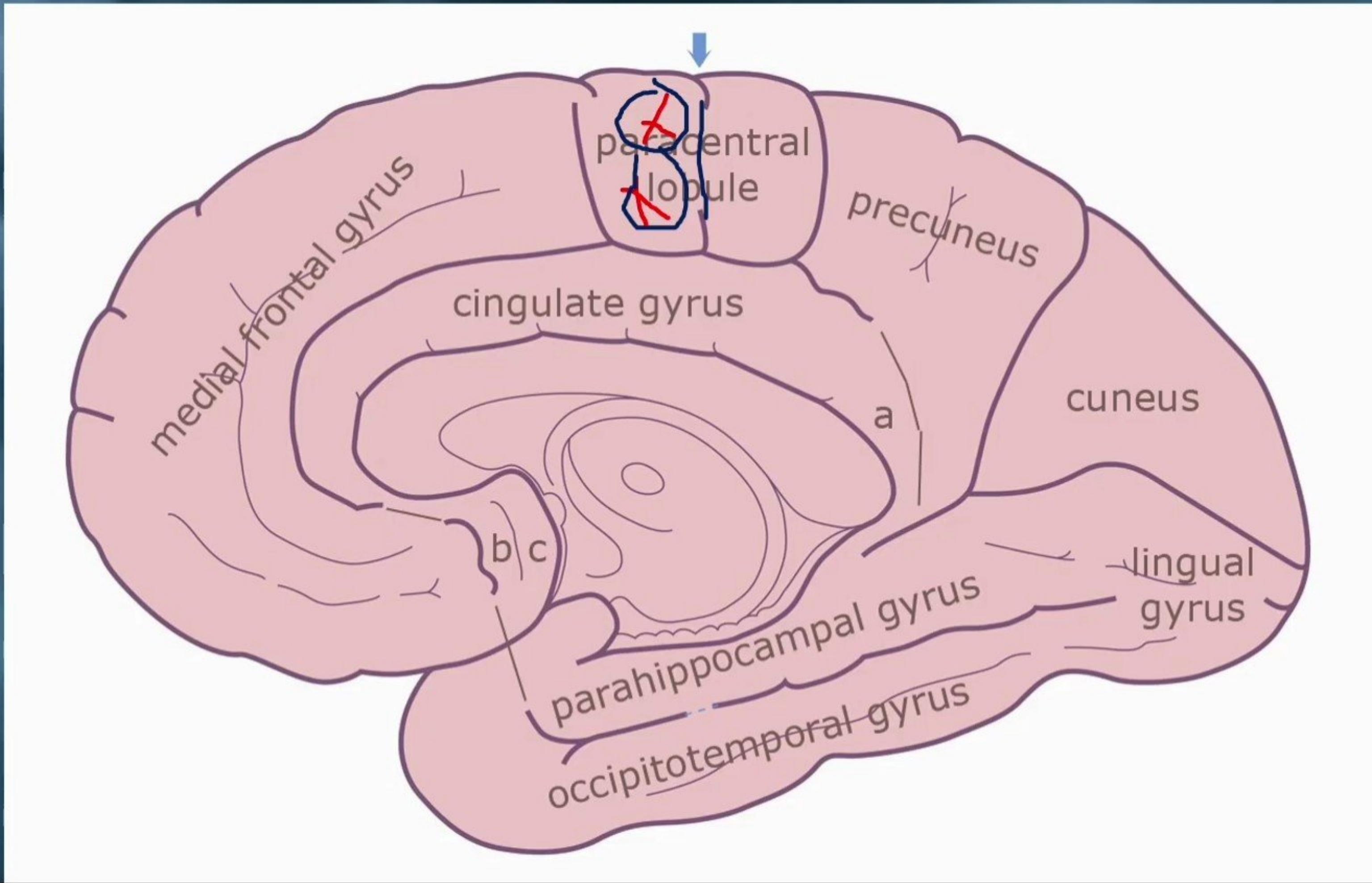
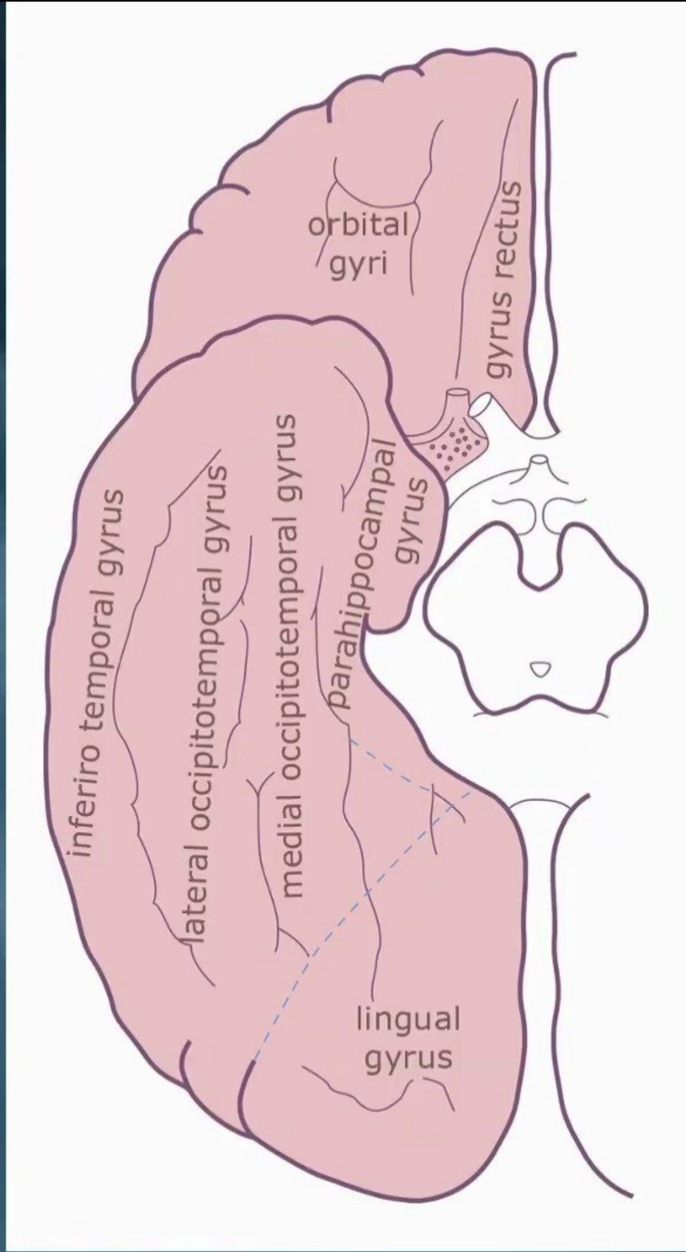


# Gyri of the Frontal Lobe

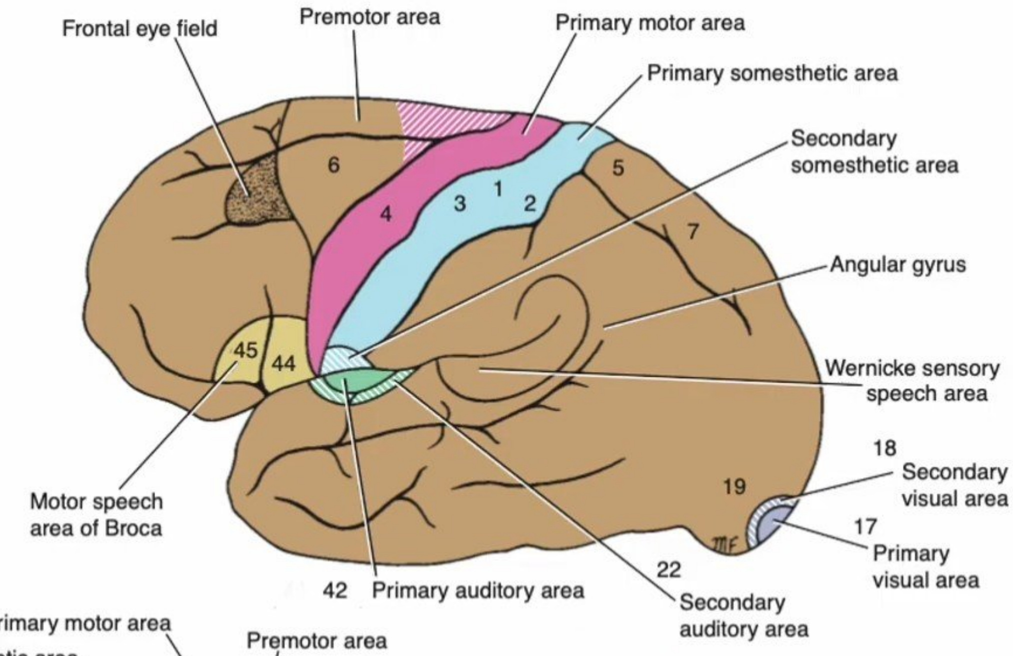
**Basal Surface**

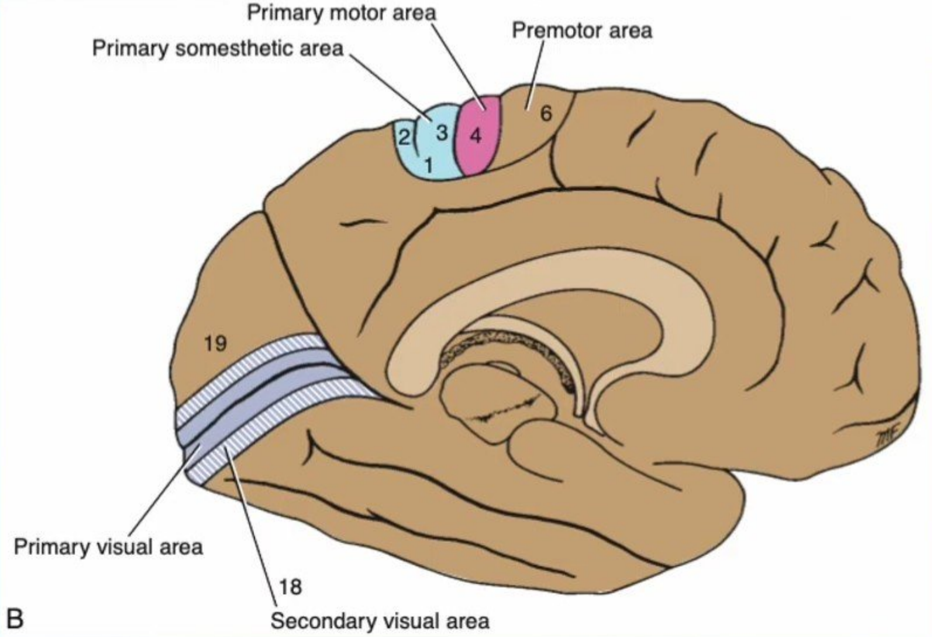
- ➢ Rectus Gyrus.
- ➢ Orbital Gyrus.



# Frontal Lobe - Functional Areas

- **Primary Motor Cortex (Precentral Gyrus):**
  - ✓ Cortical site involved with controlling movements of **contralateral** limbs of the body.
- **Premotor area.**
- **The supplementary motor area**
- **Frontal eye field (area) Cortex.**
- **Broca's Area:**
  - ✓ Controls facial neurons, speech, and language comprehension. Located on ***Left*** Frontal Lobe.
  - ✓ **Broca's Aphasia** – The inability to comprehend speech, (the decreased motor ability (or inability) to speak and form words.
- **Olfactory Bulb** - Cranial Nerve I, Responsible for sensation of Smell.
- **Prefrontal cortex.**



# Lesions of the Frontal Lobe

## Lesions of the primary motor cortex:

- Lesions of the primary motor cortex in one hemisphere result in paralysis of **the contralateral extremities.**
- (convulsion) The jacksonian epileptic seizure is due to an irritative lesion of the primary motor area.

## Lesions of the frontal Eye Field

- Destructive lesions of the frontal eye field of one hemisphere cause the two eyes to deviate to the side of the lesion inability to turn the eyes to the opposite side.

## Lesions of the motor Speech Area of Broca

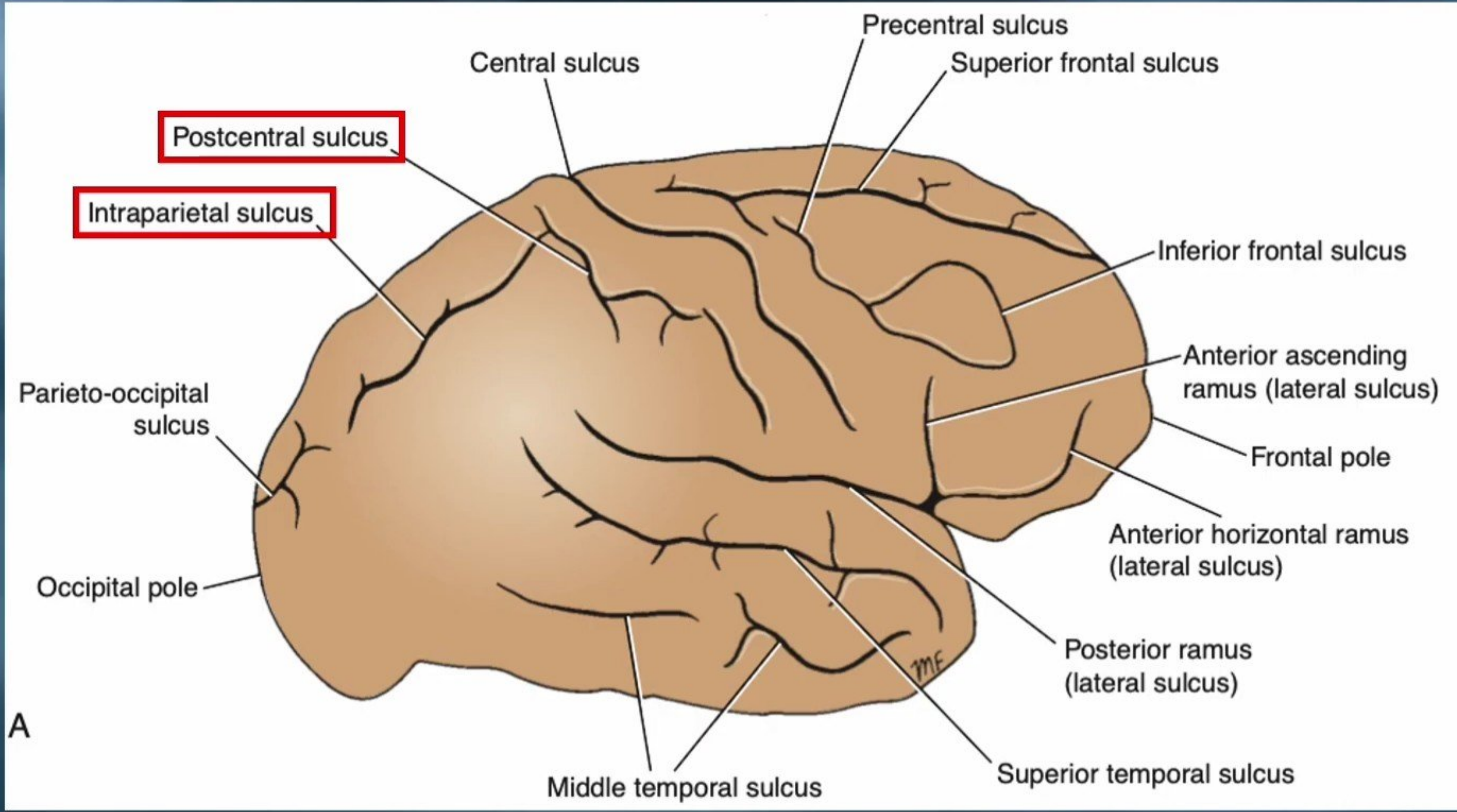
- **(Broca's Aphasia)** Destructive lesions in **Broca's area** result in the loss of ability to produce speech, that is, expressive aphasia.
- Patents can write the words, and they can understand their meaning .

## Lesions of the prefrontal cortex

- Tumors or traumatic destruction of the prefrontal cortex result in the person's losing initiative and judgment.
- Emotional changes that occur include a tendency to euphoria.
- Destruction of the prefrontal region does not produce any marked loss of intelligence.

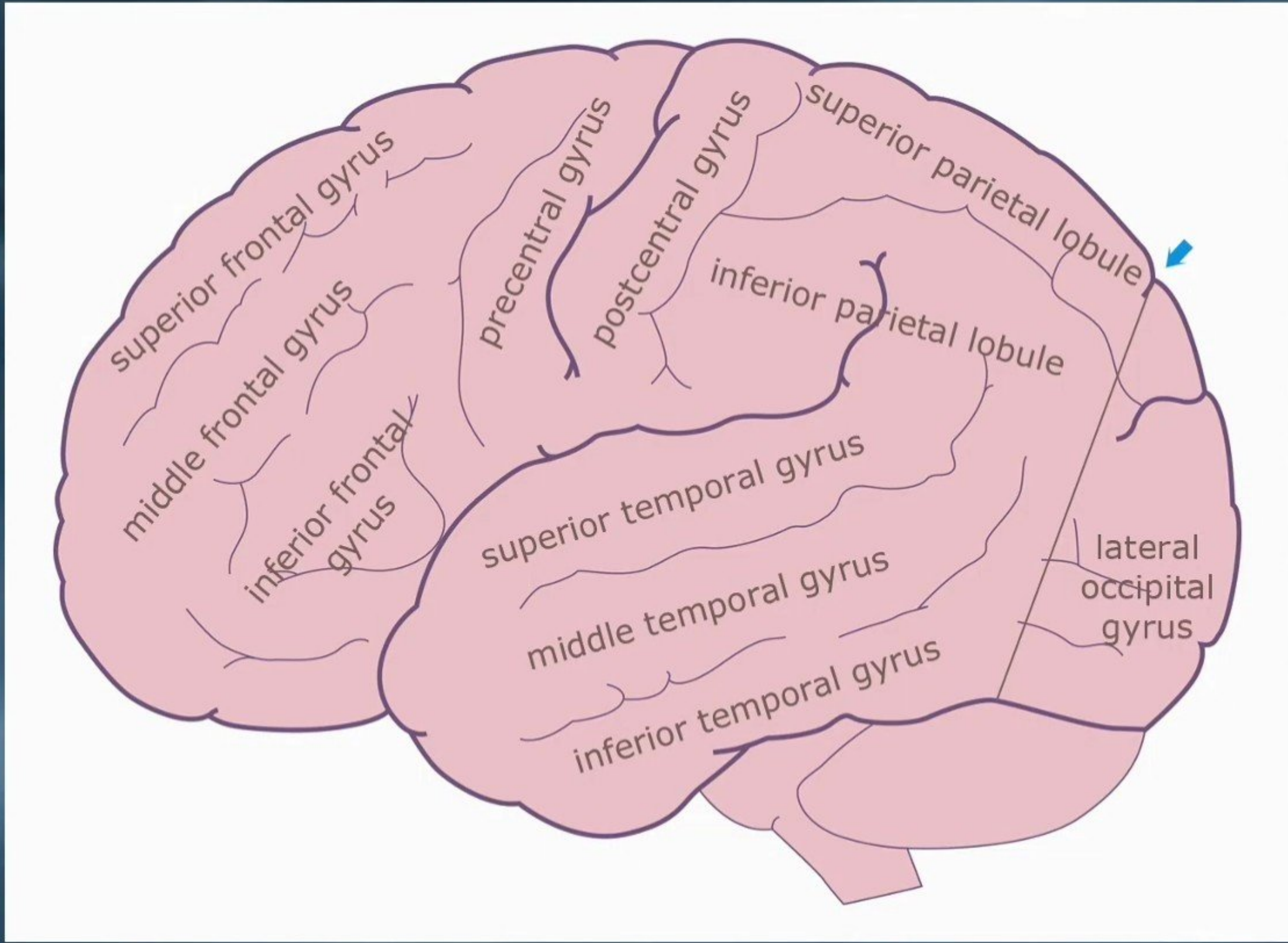
# Sulci of the Parietal Lobe

- ✓The postcentral sulcus
- ✓The intraparietal sulcus



# *Gyri of the Parietal Lobe*

- **Lateral Surface**
  - ✓ **Postcentral Gyrus - primary somesthetic area**
  - ✓ **Superior Parietal Lobule**
  - ✓ **Inferior Parietal Lobule - Wernicke's area**
    - **Supramarginal Gyrus**
    - **Angular Gyrus**



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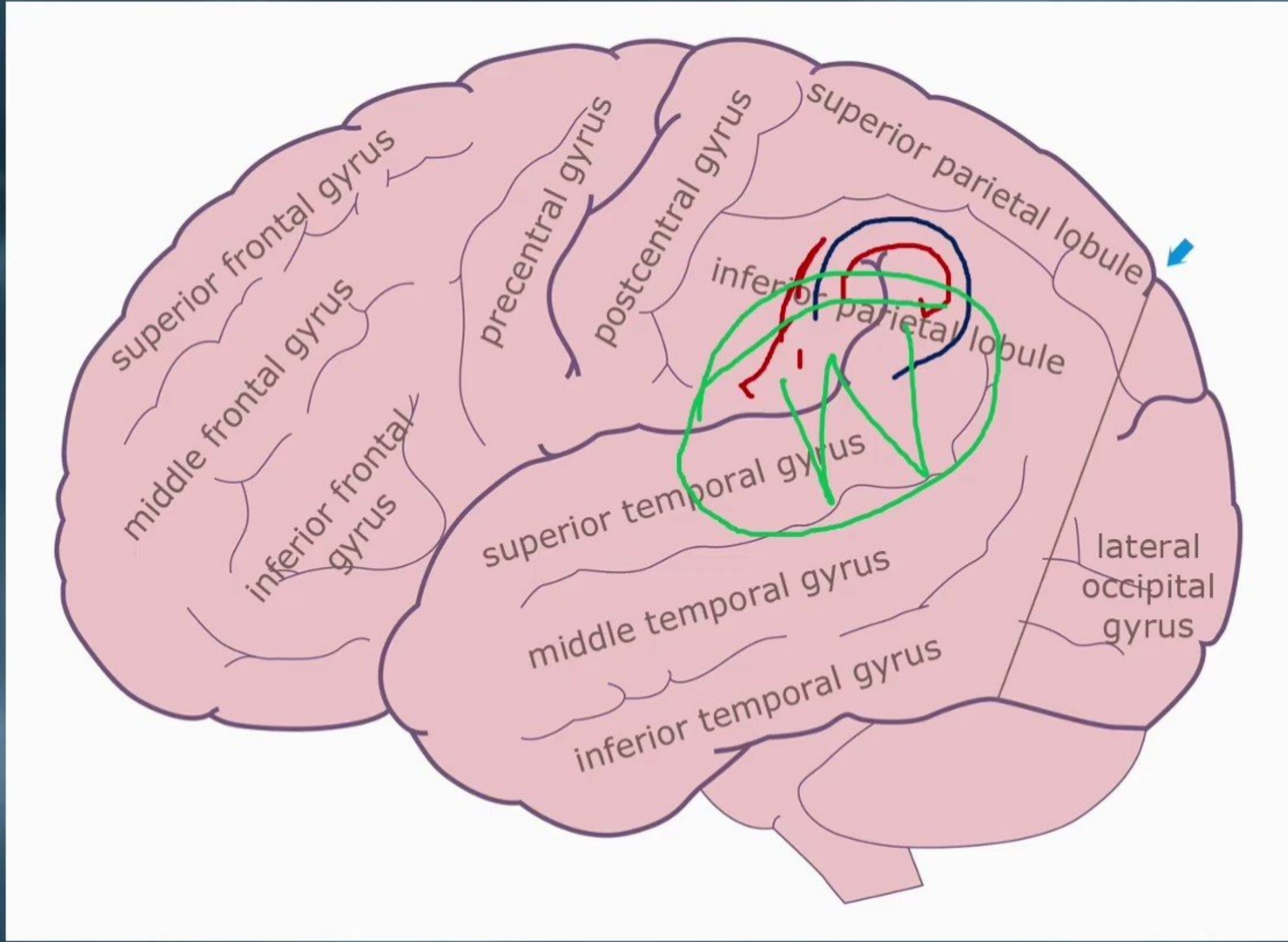
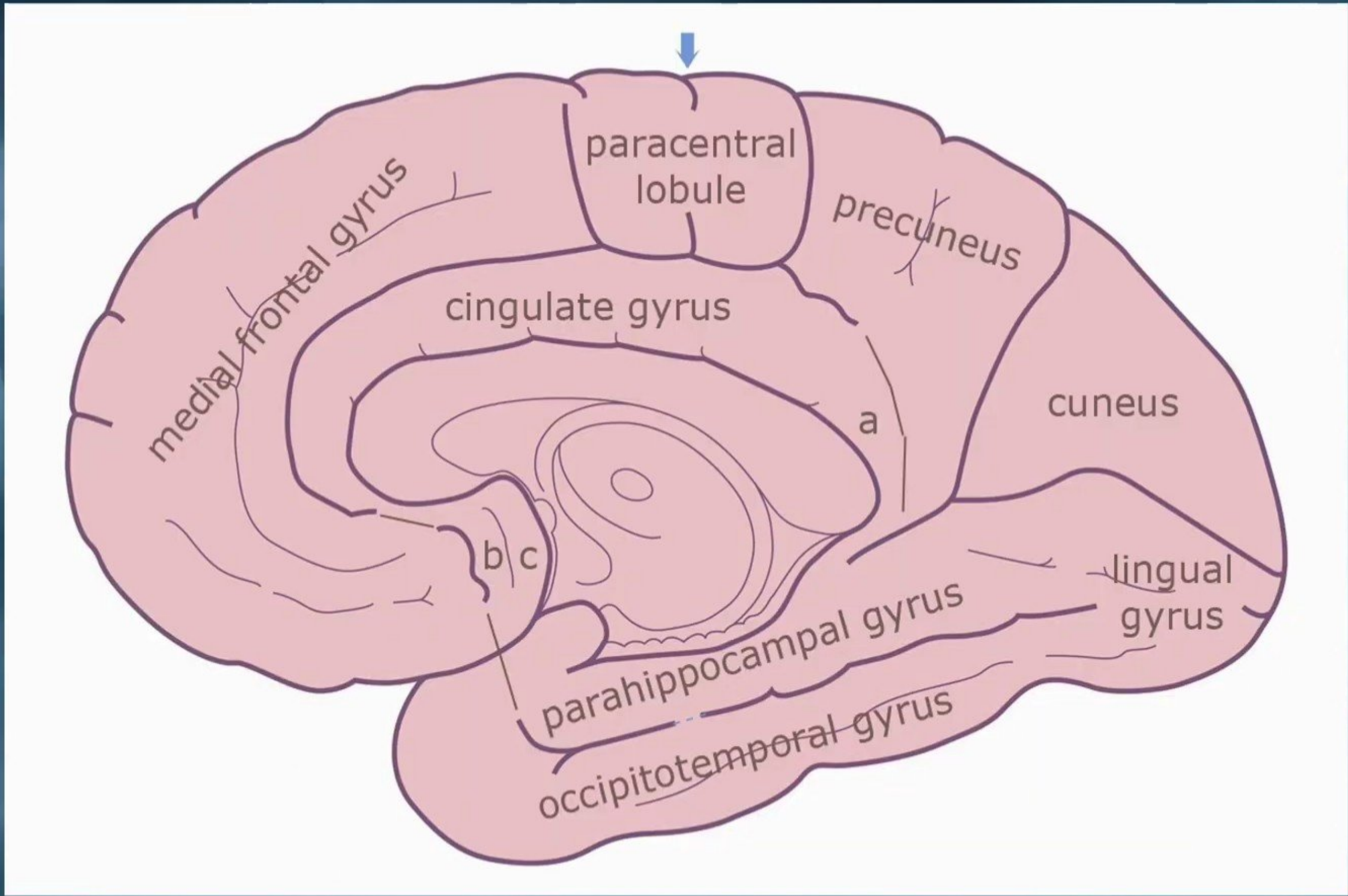


# Gyri of the Parietal Lobe

**Medial Surface**

✓ **Paracentral Lobule** - ½ sensory area

Precuneus



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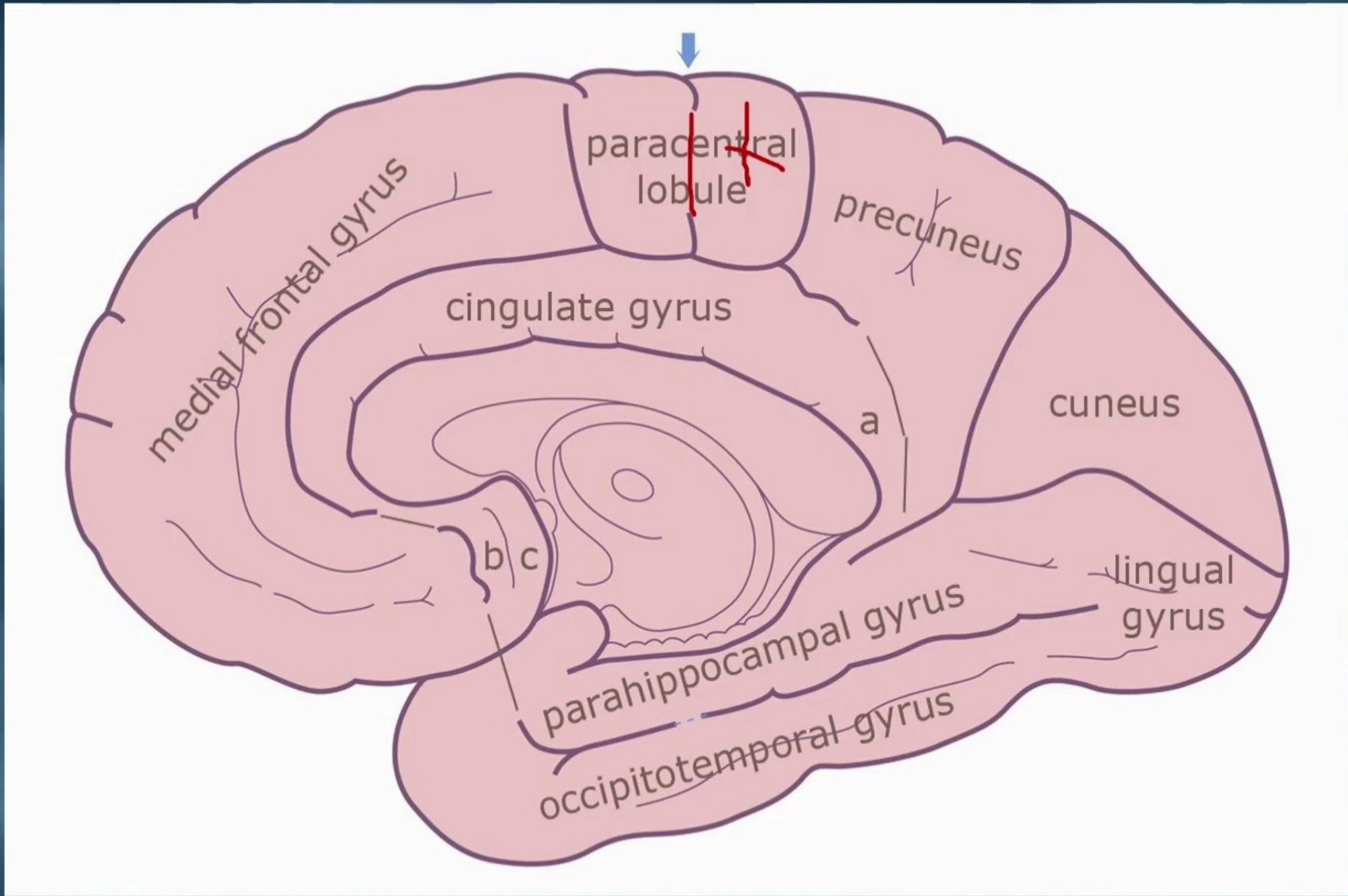
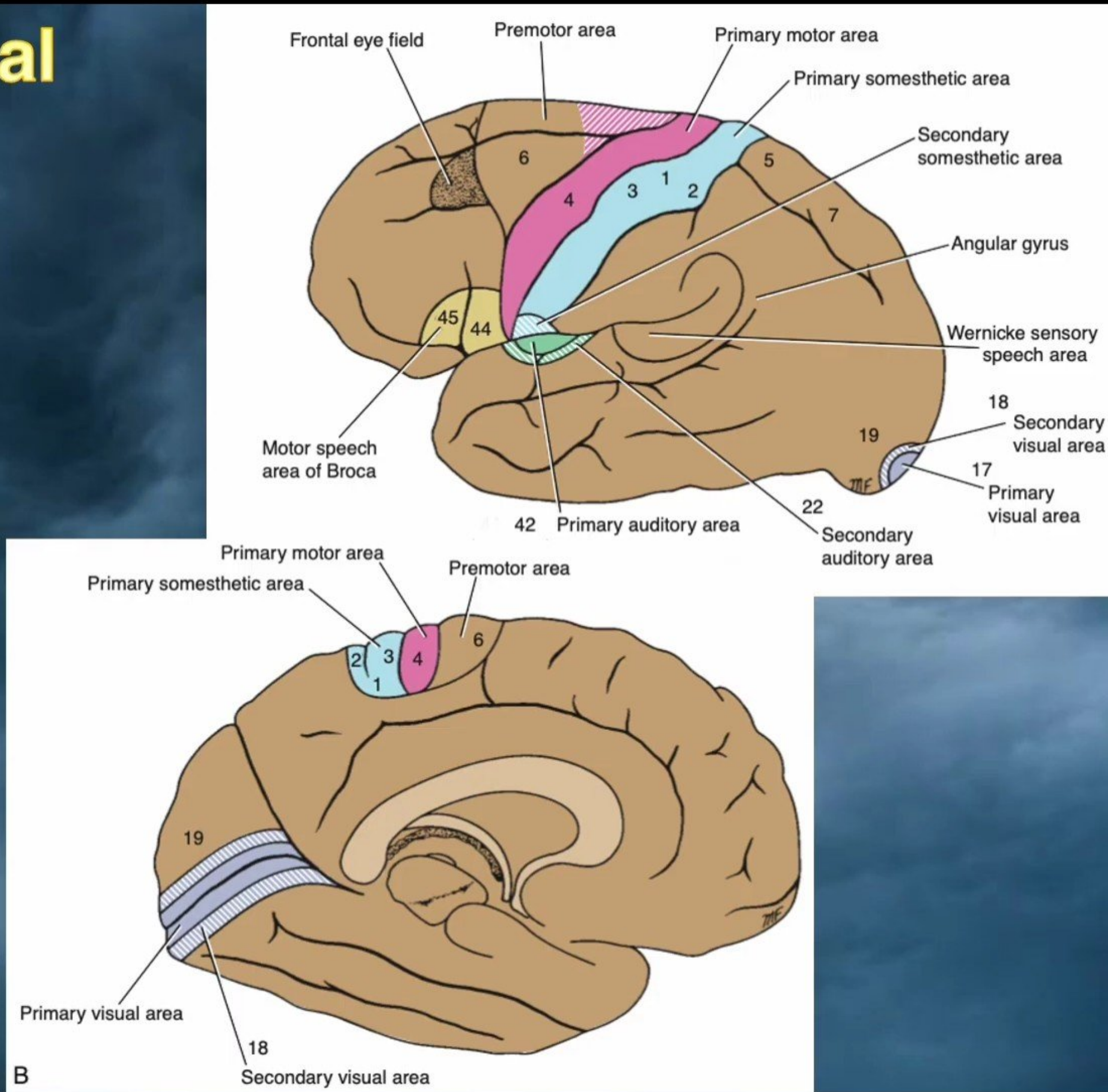


# Parietal Lobe - Functional Areas

- **The primary somatic sensory cortex (primary somesthetic area):**
  - ✓ Sensations reach the cortex from the contralateral side of the body.
- **The secondary somatic sensory cortex :** (not under-stood - tapping of the skin)
- **The association somatic sensory cortex area**
  - ✓ **It enables one to recognize objects placed in the hand without the** help of vision.
- **Primary Gustatory Cortex**
  - ✓ **Found in lower end of the postcentral gyrus and (insula).**
  - ✓ **Primary site involved with the interpretation of the sensation of Taste.**



# Lesions of Parietal Lobe

**Lesions of The primary somesthetic area**

- ✓ Lesions of the primary somesthetic area of the cortex result in contralateral sensory disturbances.
- ✓ To knows sensation was function of the thalamus.
- ✓ The patient remains unable to judge degrees of warmth, unable to localize tactile stimuli accurately, and unable to judge weights of objects.
- ✓ touch hallucinations.

**Lesions of The association somatic sensory cortex area**

- ✓ This loss of integration of sensory impulses is called astereognosis

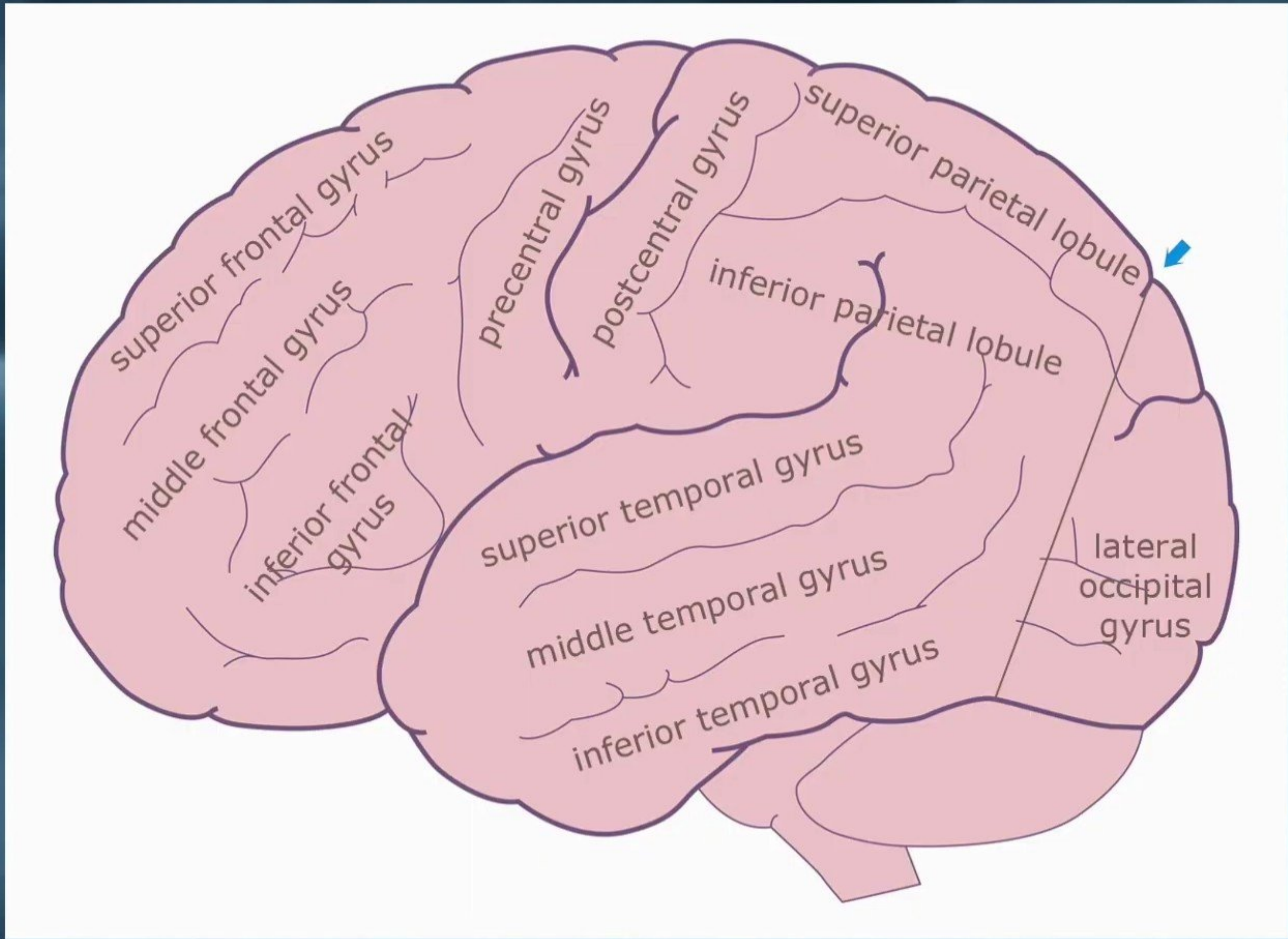
**Lesions of The Dominant Angular Gyrus**

- ✓ Destructive lesions in the angular gyrus in the posterior parietal lobe (often considered a **part of the Wernicke area**) divide the pathway between the visual association area and the anterior part of the Wernicke area.
- ✓ This results in the patient being unable to read (**alexia**) or write (**agraphia**).

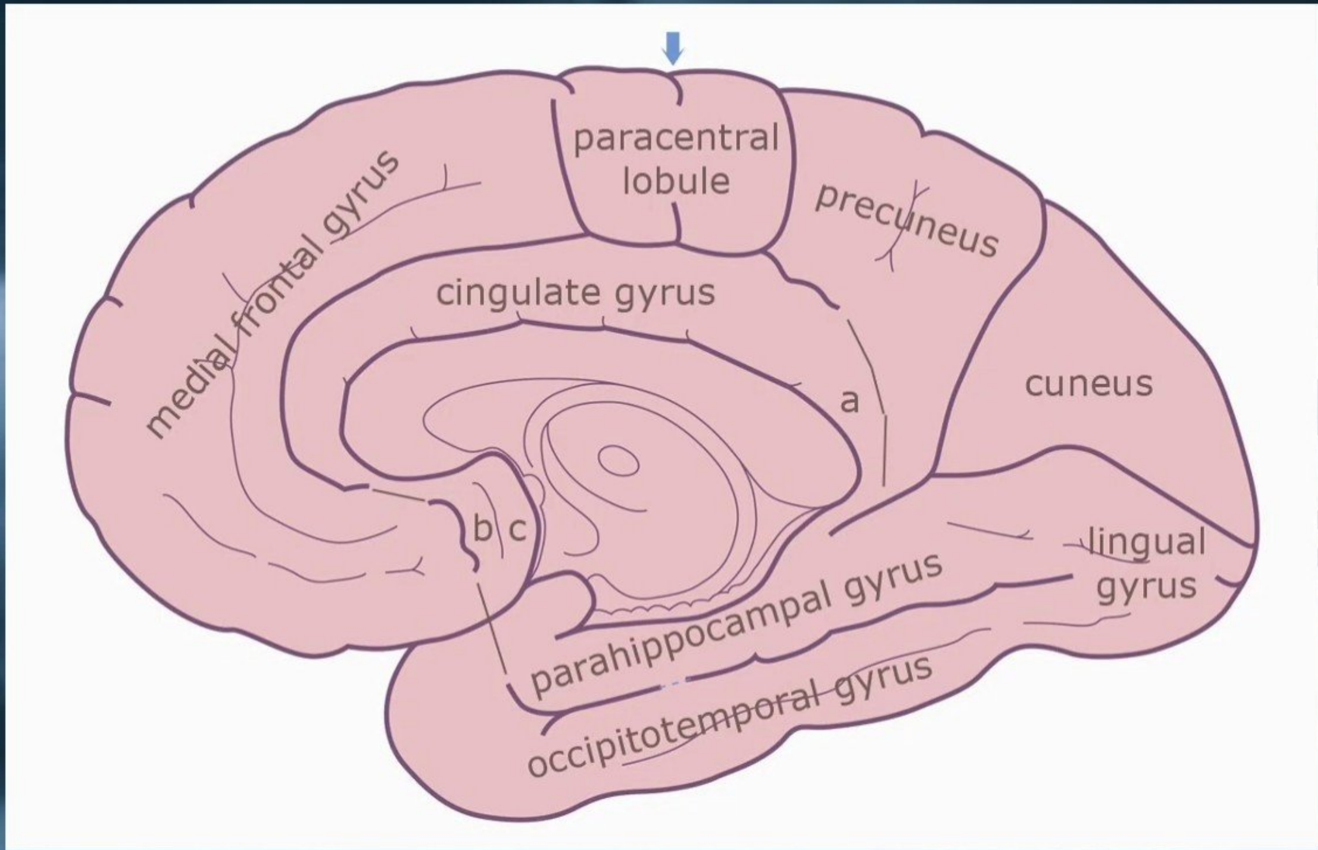
# Gyri of the Occipital Lobe

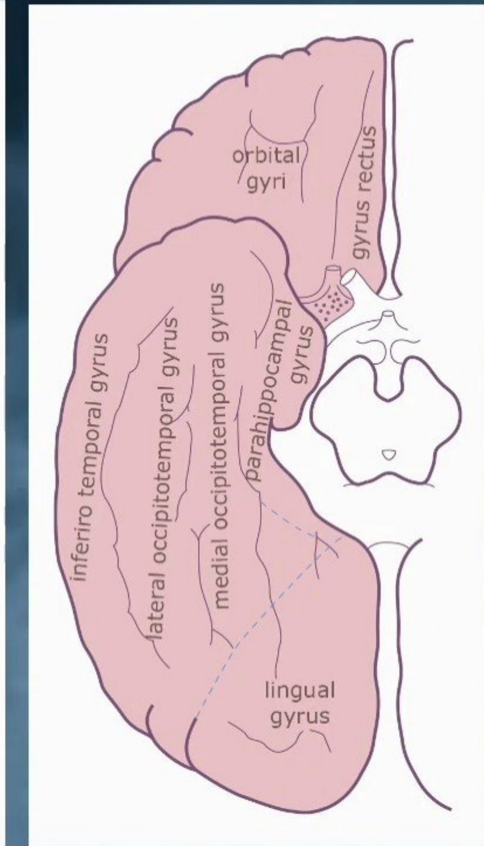
## Lateral Surface

- ✓ Lateral Occipital Gyrus



# Gyri of the Occipital Lobe



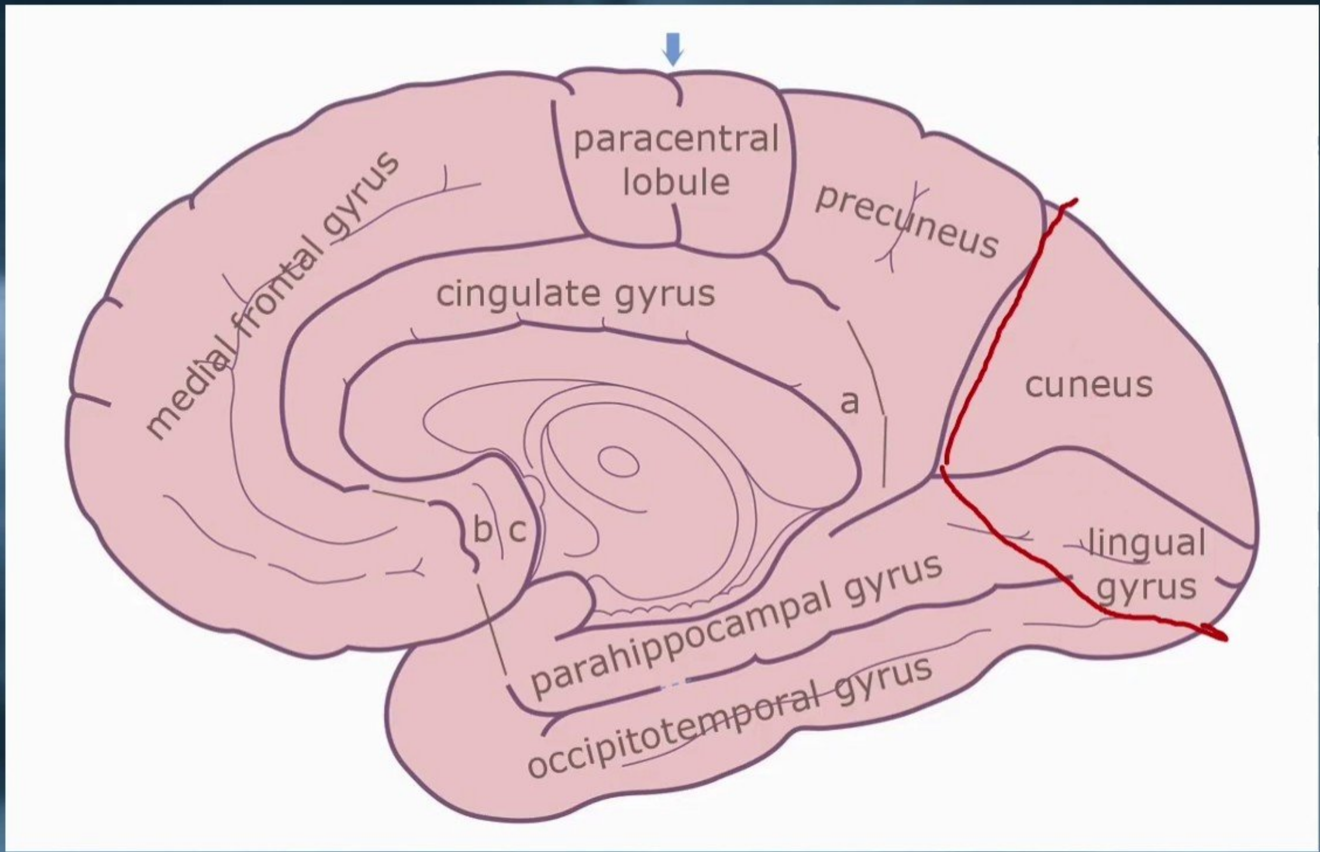

**Medial Surface**

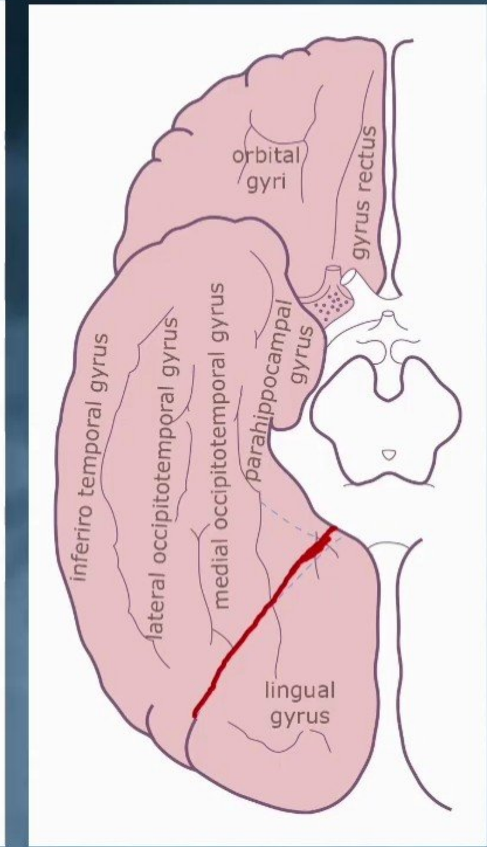
- ✓ Cuneus.
- ✓ Lingual Gyrus.

**Basal Surface**

- ✓ Lingual Gyrus

# Gyri of the Occipital Lobe




**Medial Surface**

- ✓ Cuneus.
- ✓ Lingual Gyrus.

**Basal Surface**

- ✓ Lingual Gyrus

# Occipital Lobe – Functional Areas

**The primary visual area.**

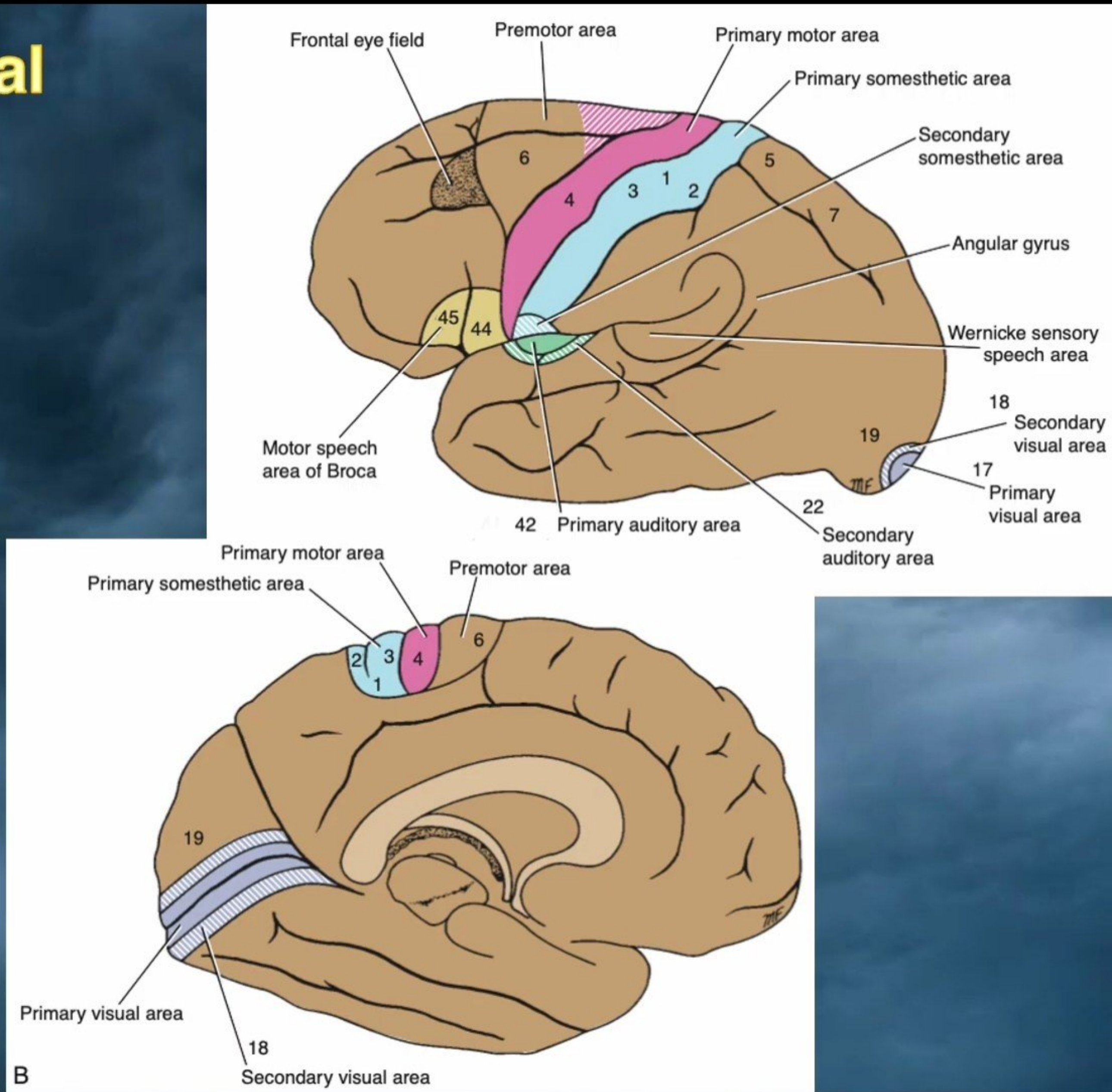
- ✓ The visual cortex receives fibers from the temporal half of the ipsilateral retina and the nasal half of the contralateral retina.
- ✓ To see.

**The secondary visual area.**

- ✓ The enabling the individual to recognize and appreciate what he or she is seeing.

**The occipital eye field.**

- ✓ It is associated with movements of the eye when it is following an object.



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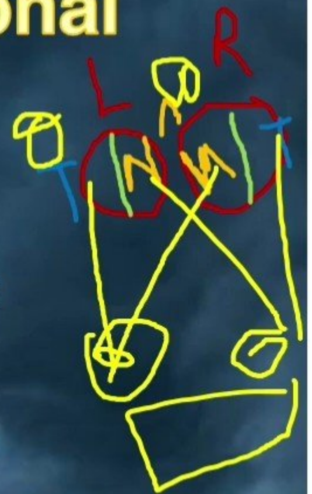

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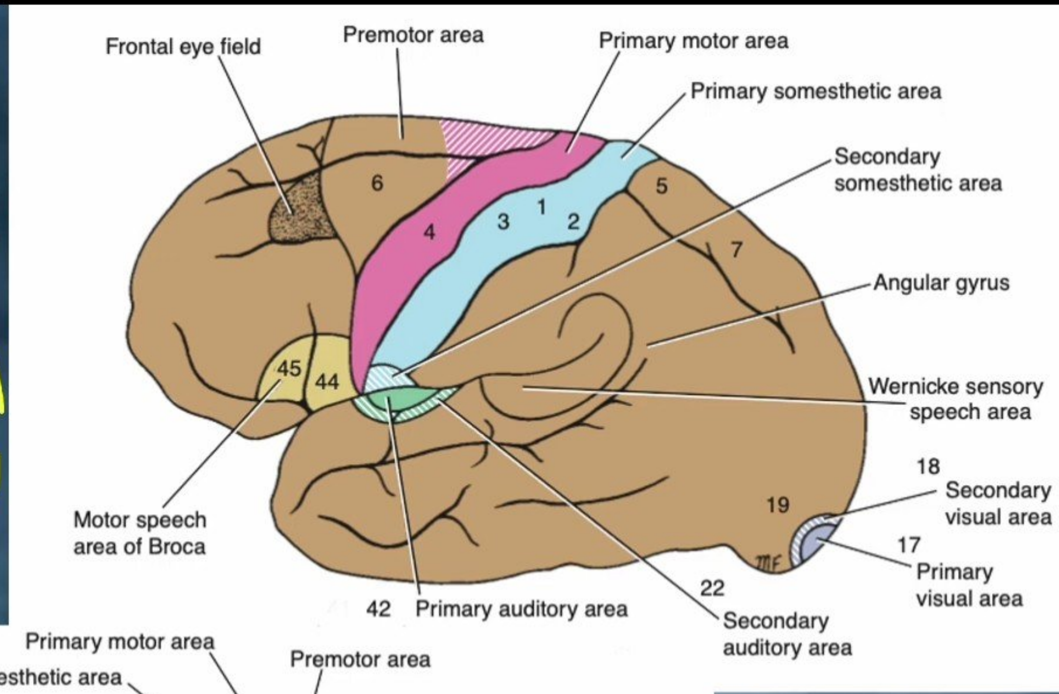
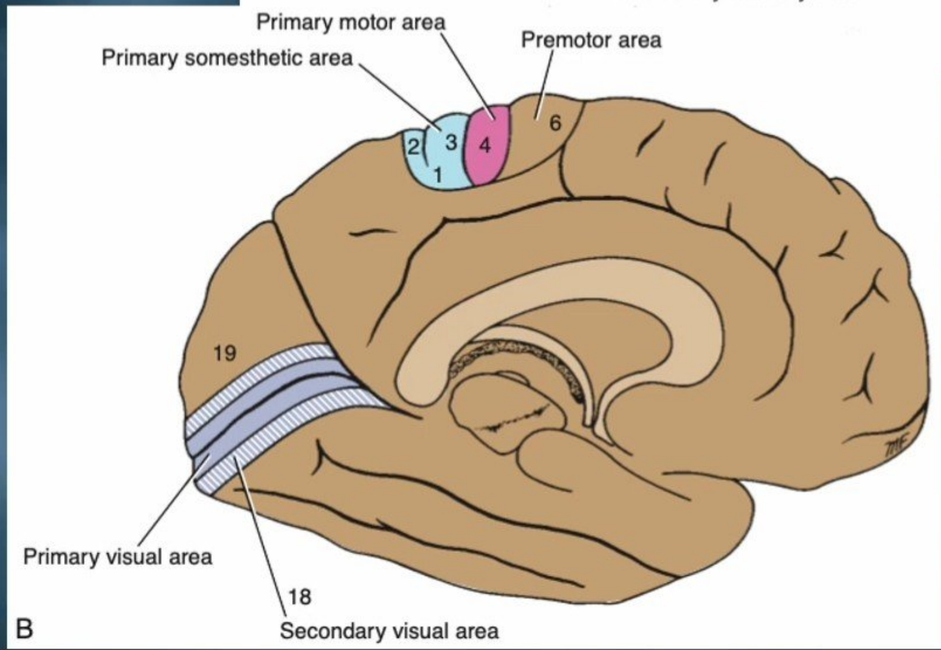
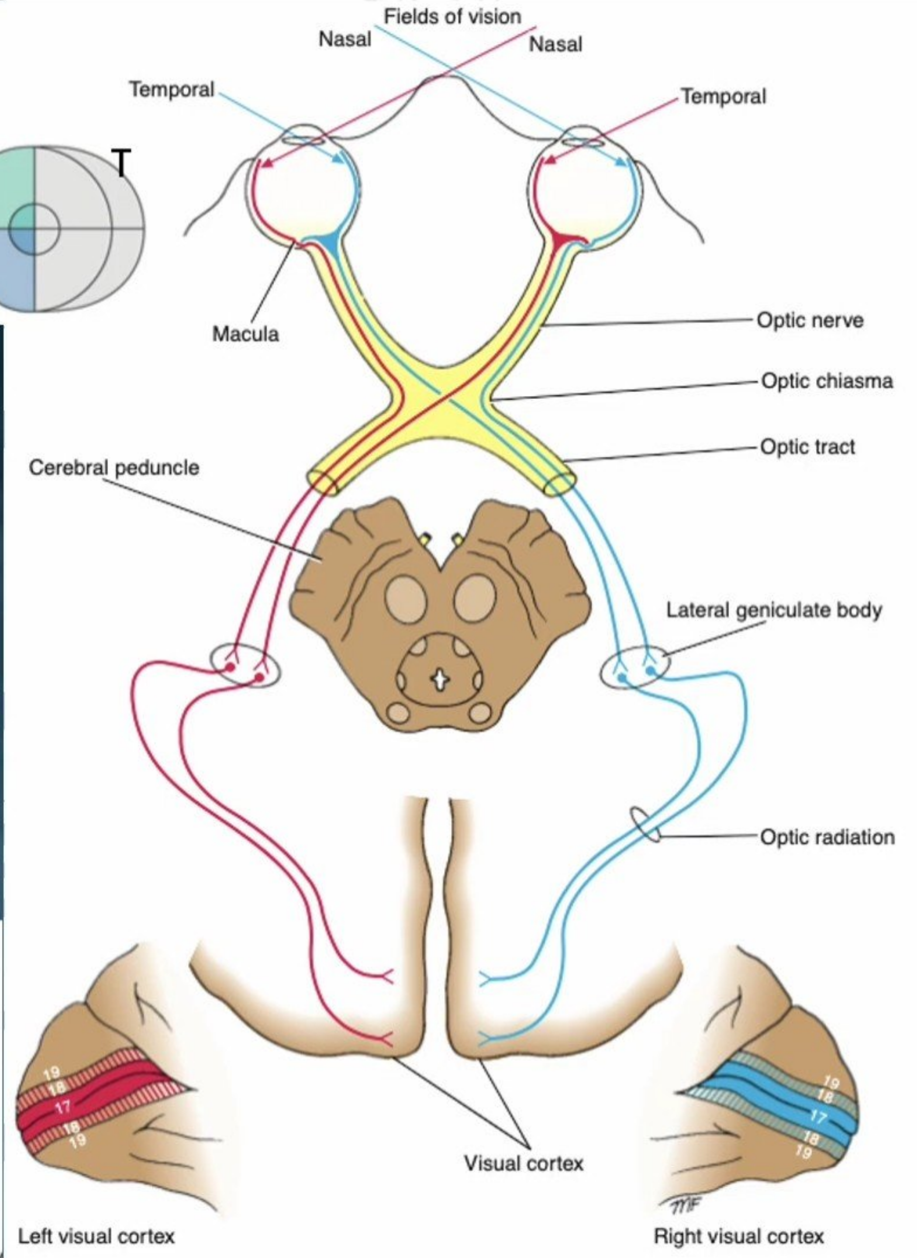


# Lesions of Occipital Lobe

Topographic organization of the visual pathways

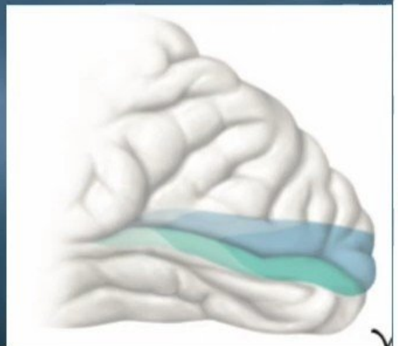




## The Primary Visual Area:

- Lesions involving the walls of **the posterior part of one calcarine sulcus** result in a loss of vision in **Left temporal** and **right nasal hemianopia** due to a lesion of **the right visual cortex**. (**homonymous hemianopia)**.
- Lesions of the upper half of one primary visual area—the **area above the calcarine sulcus**—result in **inferior quadrantic hemianopia**, whereas lesions involving one visual area **below the calcarine sulcus** result in **superior quadrantic hemianopia**.

## The Secondary Visual Area:

- Lesions of the secondary visual area result in a **loss of ability to recognize objects seen** in the opposite field of vision.



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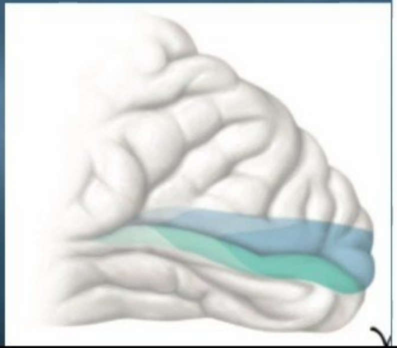

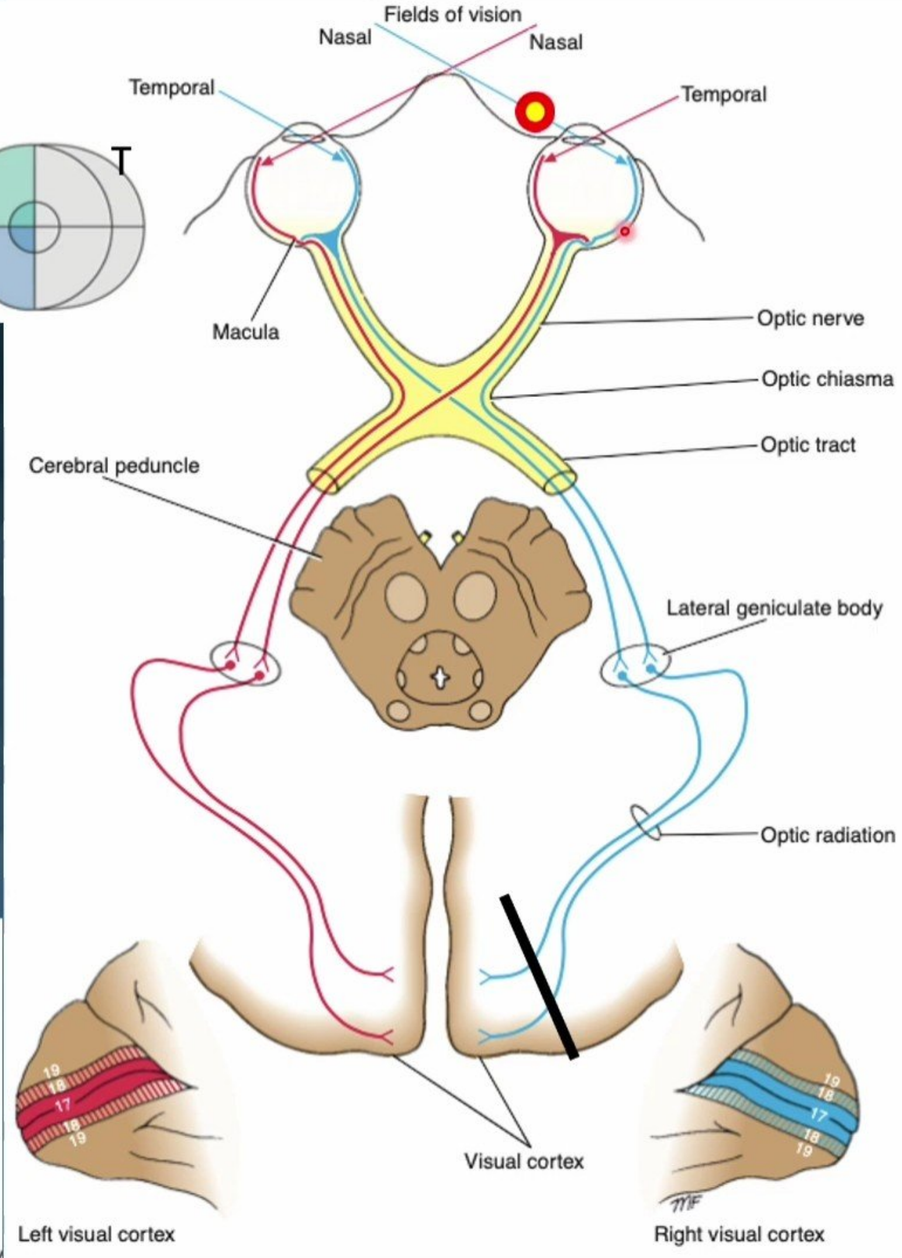

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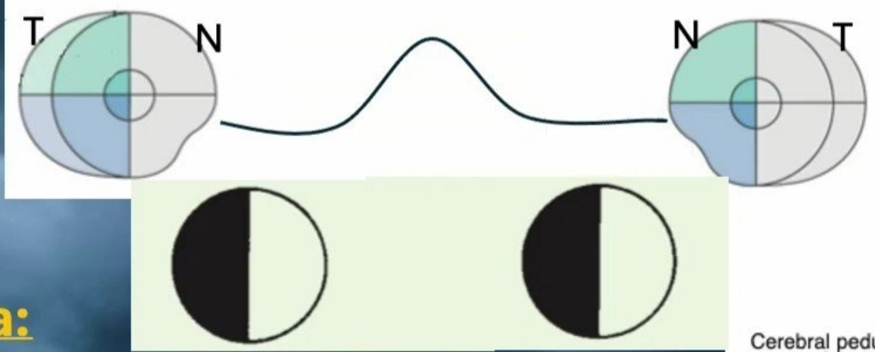
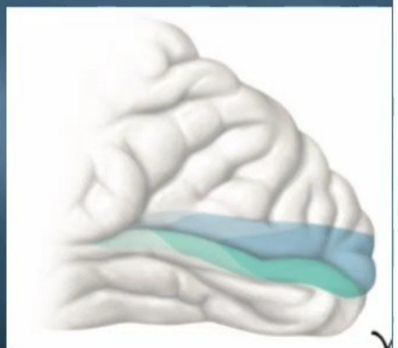
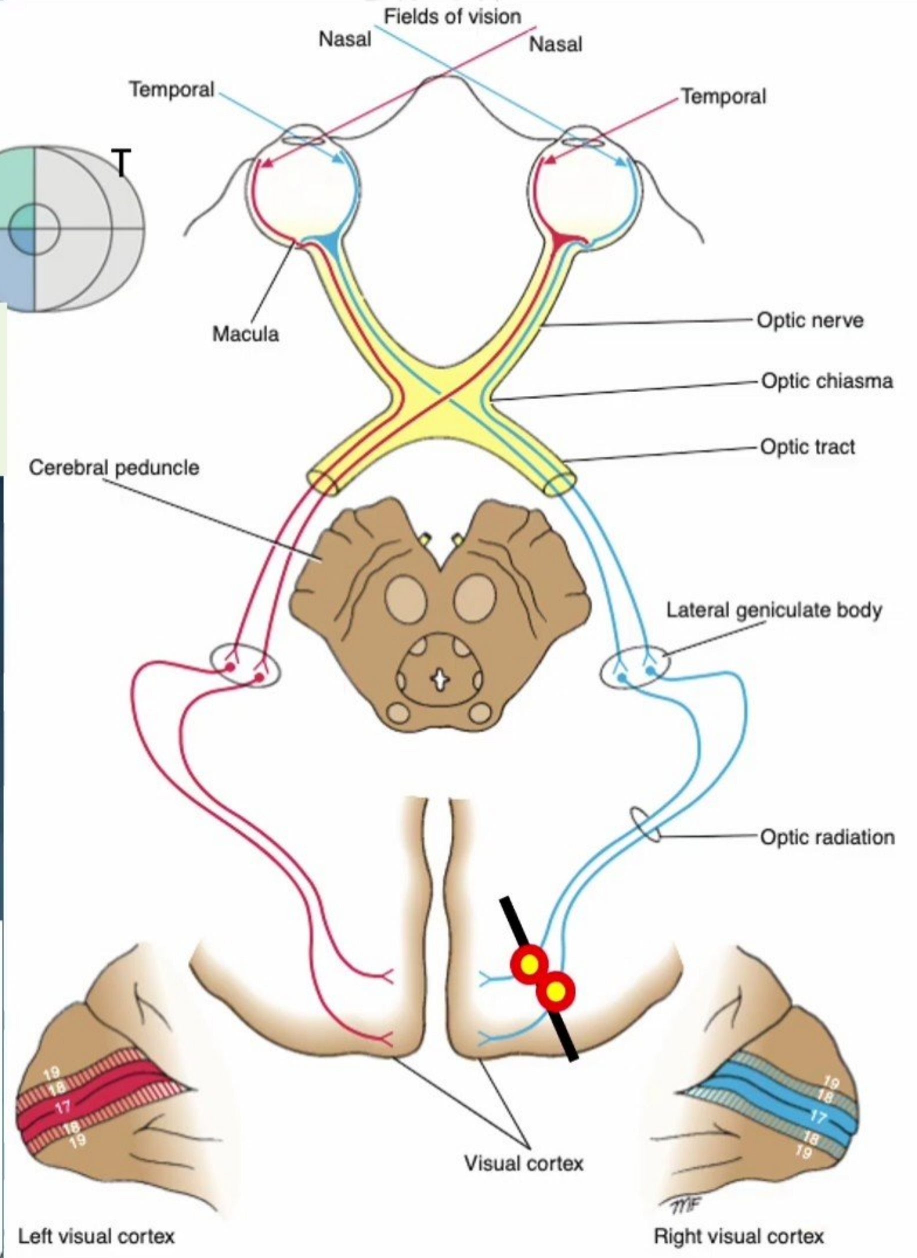


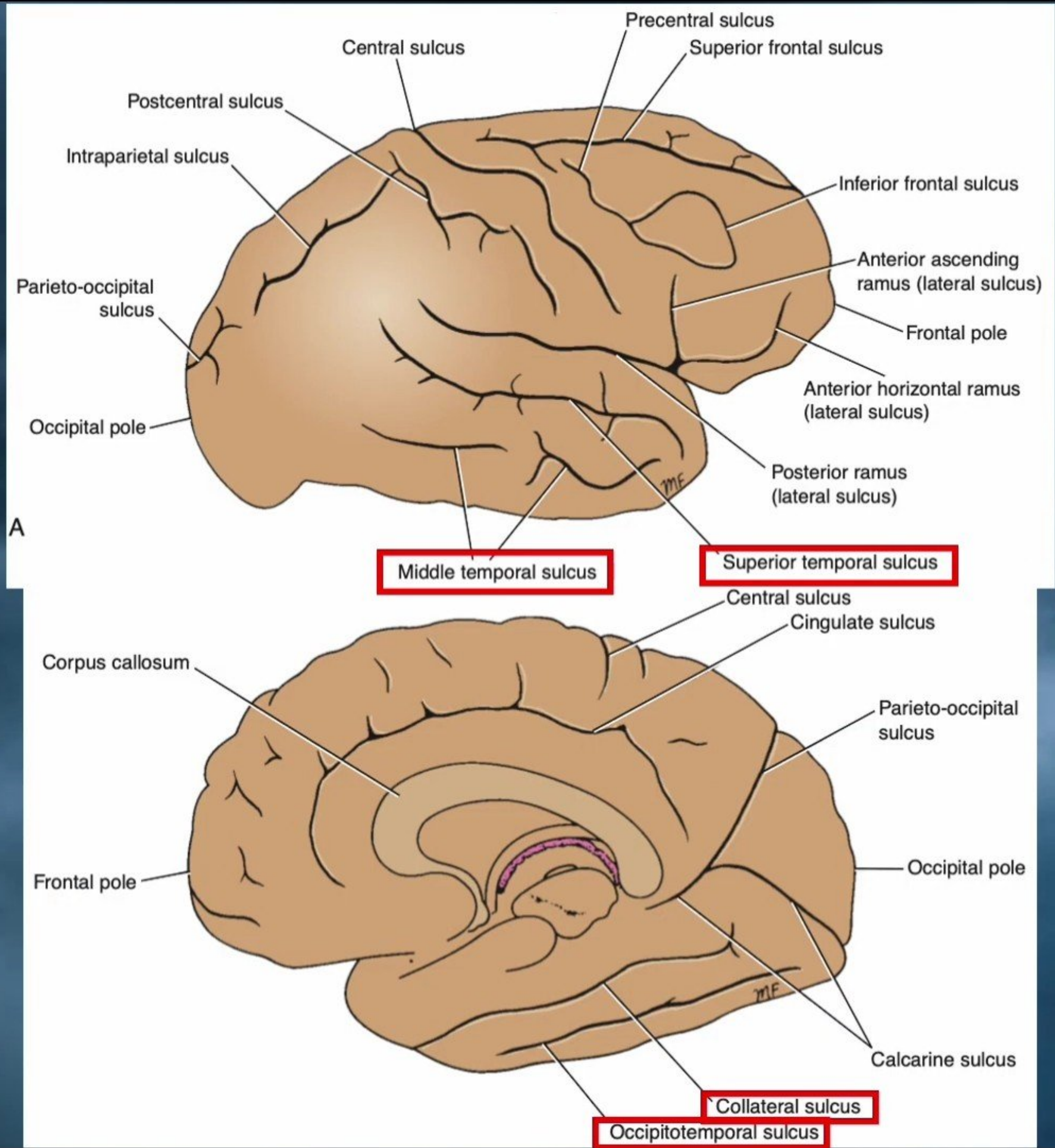





# Sulci of the Temporal Lobe

<u>In lateral view:</u>
- ✓ The superior temporal sulcus.
- ✓ The middle temporal sulcus.

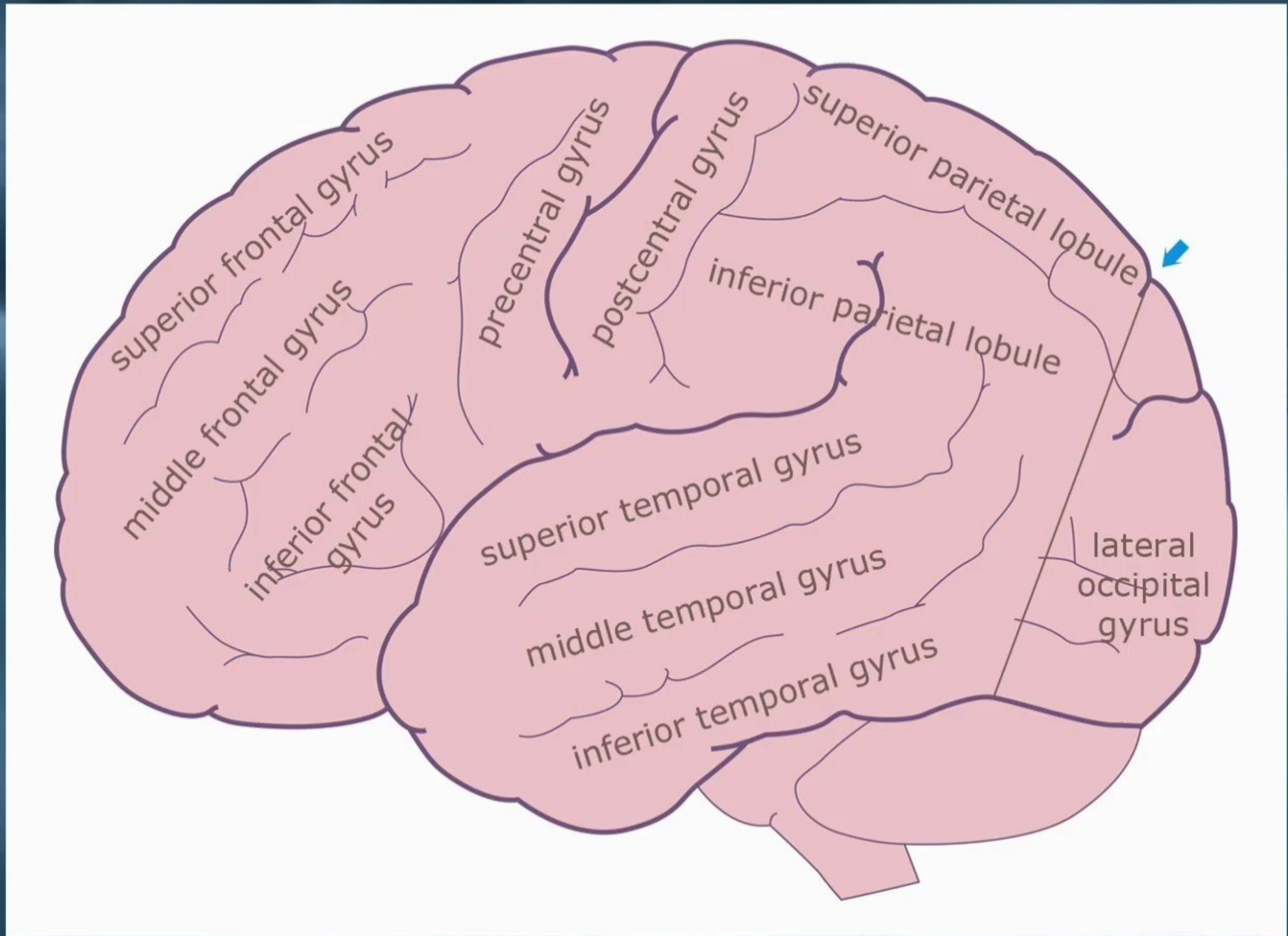
<u>In medial view:</u>
- ✓ Collateral sulcus.
- ✓ Occipitotemporal sulcus.



# Gyri of the Temporal Lobe

**Lateral Surface**

- ✓**Superior Temporal Gyrus** – auditory area
- ✓**Middle Temporal Gyrus**
- ✓Inferior Temporal Gyrus



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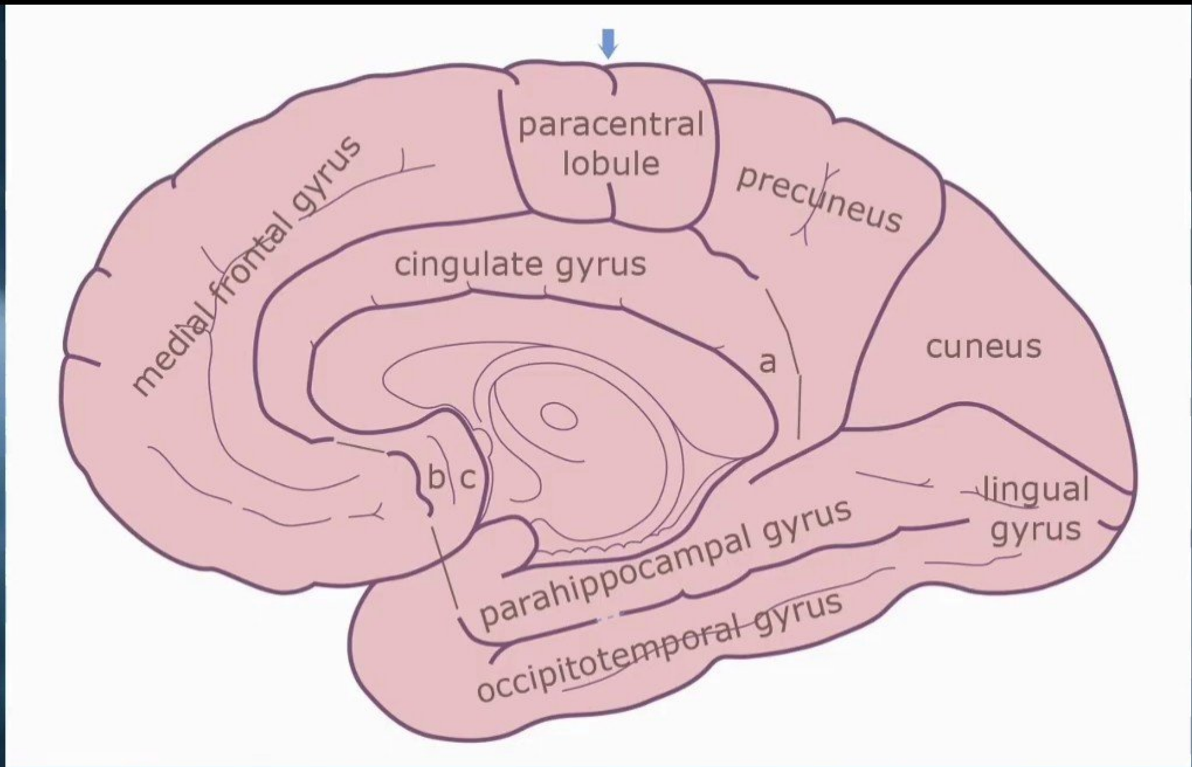
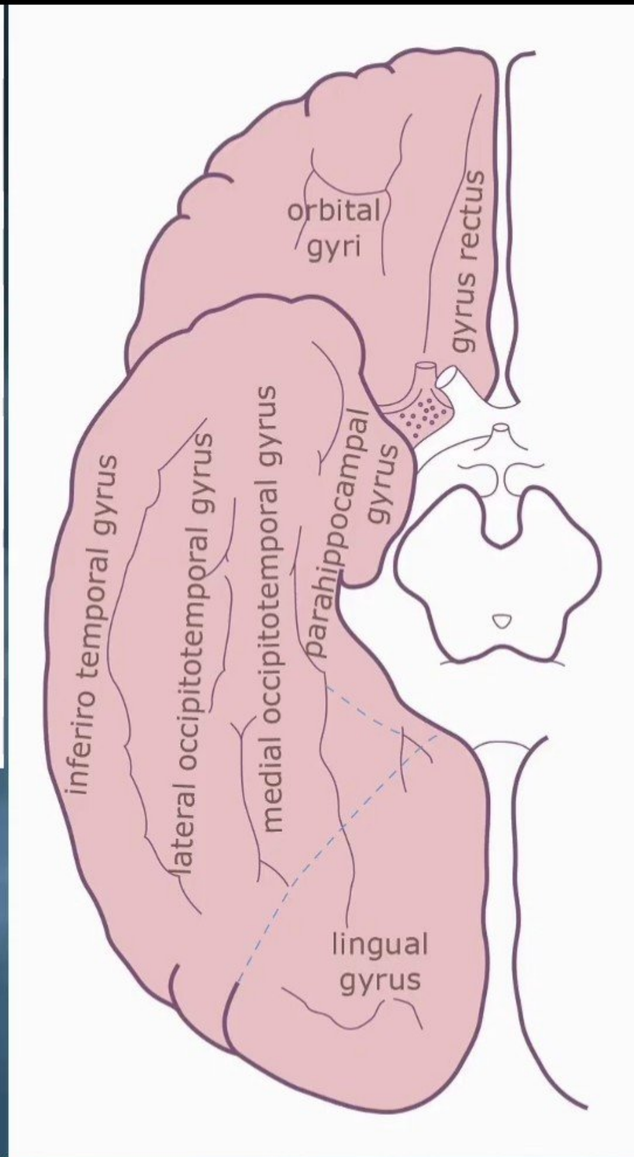



**<u>Surface Medial  and Basal:</u>**

- ✓Parahippocampal Gyrus.
- ✓Medial Occipitotemporal Gyrus.
- ✓Lateral Occipitotemporal Gyrus.
- ✓Inferirotemporal Gyrus.

# Temporal Lobe – Functional Areas

**The primary auditory area:**

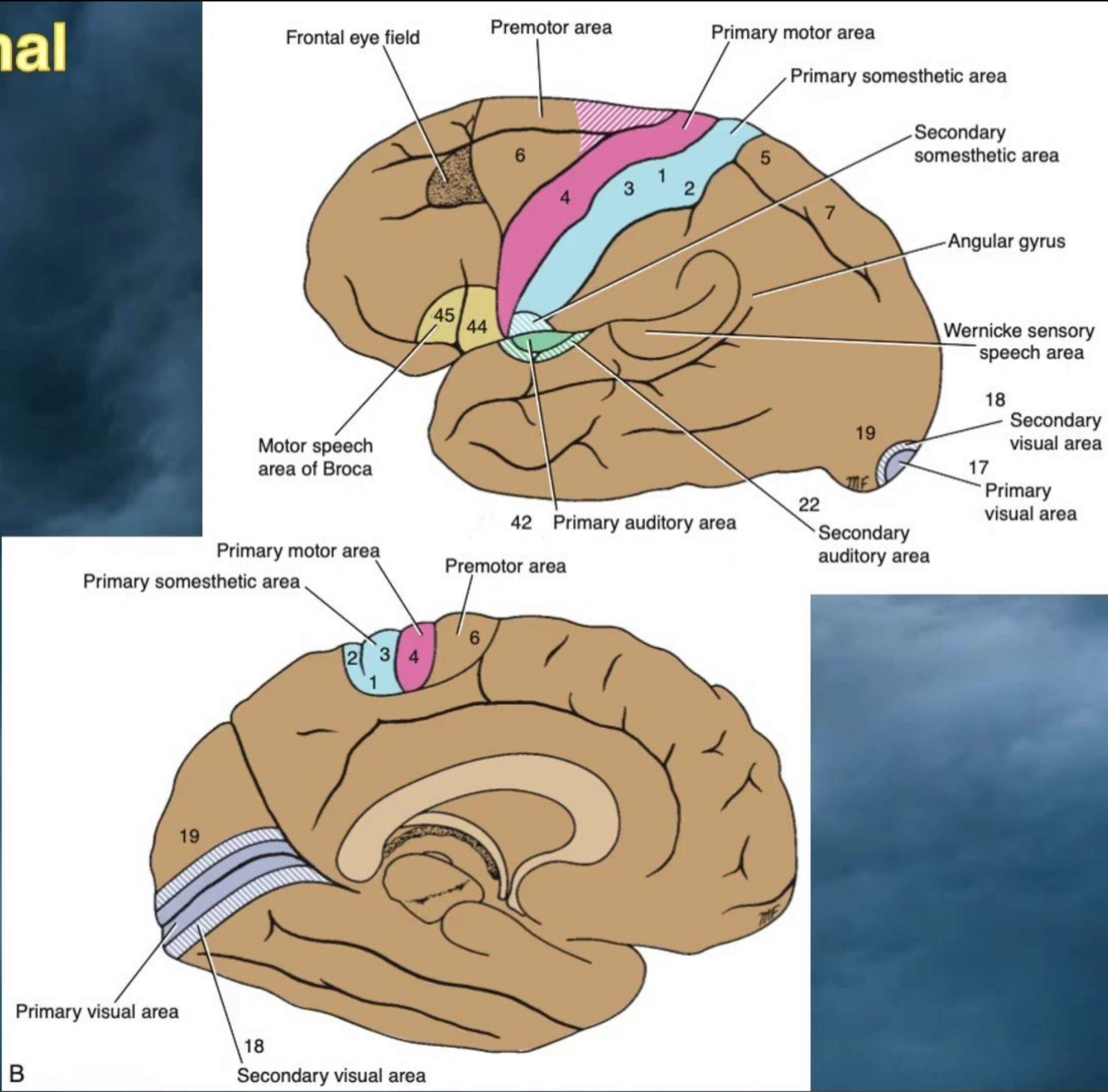
- It receives afferent fibers from **the medial geniculate body**.
- It necessary for tonal and sequential sound pattern discrimination.
- It required for **spatial localization**.

**The secondary auditory area (auditory association cortex):**

- It is to be necessary for the **interpretation of sounds** and for the association of the auditory input with **other sensory information**.

**The sensory speech area of Wernicke:**

- It is localized in the **left dominant hemisphere**,
- It permits the **understanding of the written** and **spoken language** and enables a person to **read** a sentence, **understand** it, and **say** it.
- Wernicke area is connected to the Broca area by a bundle of nerve fibers called **the arcuate fasciculus**.



# Lesions of Temporal Lobe

**The lesion of Primary Auditory Area:**

- ✓The lesion of one cortical area will produce slight bilateral loss of hearing, but the loss will be greater in the opposite ear.
- ✓The main defect noted is a loss of ability to locate the source of the sound.
- ✓Bilateral destruction of the primary auditory areas causes complete deafness.

**The lesion of Secondary Auditory Area:**

- ✓Lesions of the secondary auditory area result in an inability to interpret sounds.
- ✓The patient may experience **word deafness** (acoustic verbal agnosia).

**The lesion of Sensory Speech Area of Wernicke:**

- ✓Destructive lesions to the Wernicke speech area in the dominant hemisphere produce a loss of ability to understand the spoken and written word (**receptive aphasia**).
- ✓The patient is **unaware of the meaning** of the words he or she uses or (**uses incorrect words** or even nonexistent words). (The patient is also **unaware of any mistakes**).

**The lesion of Motor and Sensory Speech Areas:**

- ✓Destructive lesions involving both **the Broca** and **Wernicke speech areas** result in loss of the production of speech and the understanding of the spoken and written word (**global aphasia**).
- ✓Patients with lesions in **the insula** have **difficulty in pronouncing phonemes (**produce sounds to the target word but are not exactly correct).

Will be continue ...